EU-SILC · EUROSTAT · 2003–2025

The Greek Poverty Paradox

Greece's official income-poverty rate is elevated but not extreme among EU countries — 4th-highest of 27. A broader official measure, AROPE, is also elevated. But by their own account, Greek households are the most financially squeezed in the Union by a much wider margin, and have been for over a decade. This report investigates how much of that gap is real.

A note on framing: this project started from the question "objective vs. subjective poverty," but that shortcut is too loose. This report uses **AROP** for the EU's relative-income poverty line (60% of national median income), especially when explaining the moving-threshold problem; it uses **AROPE** for the broader official poverty-or-social-exclusion comparison; and it uses **subjective poverty** for households saying they struggle to make ends meet. None of the official measures is treated as ground truth against which households are being checked.

EXECUTIVE SUMMARY

The argument runs in six movements: the AROP puzzle and the EU's own broader measure as a partial bridge; why AROP's ruler shrank during the crisis; how broadly Greek conditions deteriorated; a model built to exclude anything resembling the answer; what that model does *not* establish, including two designs that were built and failed; and what follows for reading AROP in a crisis.

The AROP puzzle. Two out of three Greek households say they struggle to make ends meet — the highest share anywhere in the EU, more than triple the EU average, every year since 2011. Greece's official income-poverty rate, AROP, tells a much less dramatic story: 4th-highest of 27, elevated but not an outlier. The raw difference between the two is 47.6 points, more than three times the next-largest in the EU. This report treats that as a puzzle to be examined, not as households being wrong.

The shrinking ruler. Part of the puzzle is arithmetic. AROP's line is not a fixed income level; it is 60% of whatever the national median happens to be that year. When Greek incomes collapsed together, the line collapsed with them, so AROP barely moved while households got dramatically poorer underneath it. Held to a fixed pre-crisis standard, measured poverty roughly doubles during the crisis, closely tracking what households reported.

The AROPE bridge. AROPE — income poverty *or* severe deprivation *or* very low work intensity — is the official system's own recognition that income poverty alone is

too narrow. It narrows Greece's raw difference to 39.7 points and does not close it. AROPE motivates what follows; it does not replace AROP as the object.

How broad the deterioration was. Before any model: across 25 indicators chosen to exclude the outcome and every model variable, the share placing Greece in the EU's worst fifth rose from 21% before the crisis to 58% from 2012, and 68% in 2024 — first of 27. Real wages remain further below their own 2008 level than any other member state's. This is descriptive evidence of how widely the collapse reached, and it is not used as an explanation.

The central result. A model was built to exclude anything that restates the outcome — no arrears, no inability to meet an unexpected expense, no expectations measures. Fitted on the other 26 countries and used to predict Greece, it leaves Greece **27.05 points** under-predicted, the largest gap in the Union. Adding the accumulated record of unemployment since 2009 brings that to **+6.93 points**, moving Greece from 1st to **3rd of 27**. That is the report's headline, and it is a comparison between two models rather than a decomposition of the raw AROP difference.

What that does not establish. The association is supported **predominantly between countries**: countries carrying more accumulated exposure report more hardship. The panel does not show that as Greece accumulated more exposure its own hardship rose, and its within-country estimate is too imprecise to rule that out either way. Greece remains the third most under-predicted country in the Union: roughly seven points are still unaccounted for. Two alternative explanations were pre-registered, built and **failed** — a synthetic-control comparison whose best-matching counterfactual had half Greece's income and three times its deprivation, and a multi-domain breadth index that moved Greece's prediction the wrong way.

On the smaller residuals in earlier versions. Earlier specifications in this project admitted arrears and unexpected-expense capacity and reported visibly smaller Greek residuals — in the richest of them, close to zero. Those results are retained in the legacy section and the statistical appendix, and they are **superseded**. Their explanatory success came from predictors that restate the outcome: a model that explains difficulty making ends meet using arrears has not explained much. The larger, more honest +6.93 is the headline.

Reporting culture. A stable Greek reporting style cannot explain why the gap widened after 2009 by more than any other member state's, nor why the extremity is concentrated in financial questions rather than spread across all self-reports. What remains open is narrower and is stated as open: a **financial-domain-specific**

reporting difference arising from the crisis itself, which no aggregate design used here can exclude.

Three different quantities, not one ladder. This report reports numbers that must not be added together or read as stages of a single decomposition:

• the **raw AROP difference** (47.6 points in 2025) — a descriptive subtraction of one published rate from another;

• the **AROPE bridge** (39.7 points) — the same subtraction against the EU's broader official measure;

• the **prediction residual** (+27.05, then +6.93) — how far Greece's reported hardship sits from what a cross-country model fitted *without* Greece predicts for it.

The first two are differences between measures. The third is a modelling quantity with a different estimand, and moving from 27.05 to 6.93 is a comparison of two model specifications — not "points of the AROP difference explained".

A note on limits. Every figure comes from official Eurostat data except one: a poverty measure anchored to 2008 living standards, which Eurostat does not publish that far back and had to be approximated (validated against a real official measure to within about 1 point on average). All results are associational, drawn from aggregate country-level data rather than household microdata. The headline is **post-selection robustness, not independent confirmation**: accumulated exposure was discovered on this same panel, so this validates a pre-specified model against the data that produced the hypothesis. Full limitation structure in Methods at the end.

Subjective poverty, 2025	AROP income poverty, 2025	AROP gap, 2025	Objective-only residual
67.2%	**19.6%**	**47.6 pp**	**+6.93 pp**
vs EU 17.6%; rank 1st	vs EU 16.3%; rank 4th	subjective poverty minus AROP; largest in EU	from +27.05 without accumulated exposure; rank 3rd of 27

The argument, in six movements

Within the movements

MOVEMENT 1

The puzzle

The question: Greek households report the EU's highest difficulty making ends meet, while the official income-poverty rate places Greece high but not alone. How large is that divergence, and does the EU's own broader measure close it?

What is being measured. The outcome throughout is the **backward-extended official subjective-hardship indicator**: Eurostat's own published series (`ilc_sbjp01`) from 2010 onward, and before 2010 the identical `DIF + GRT` construction from the underlying survey. The two were reconciled across all 432 overlapping country-years: 318 agree exactly, 114 differ by a single rounding step, and none differs by more than 0.1 points. This is not a bespoke measure competing with an official statistic — it *is* the official statistic, carried back to 2003 by the rule Eurostat itself applies from 2010. Note the limit of that validation: it establishes the aggregation rule, not that pre-2010 vintages are free of national survey breaks.

1

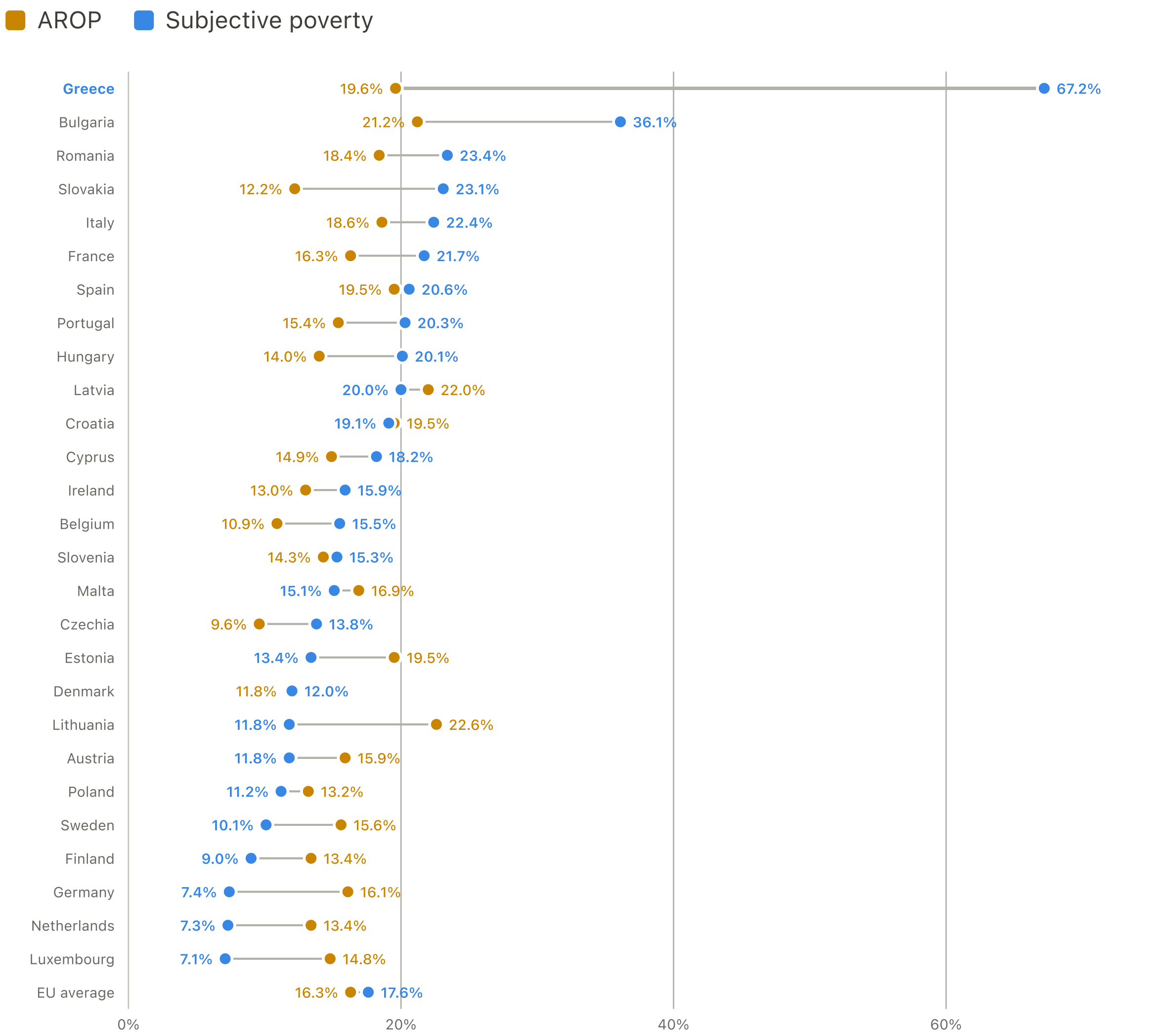
The paradox: subjective poverty vs. official measures

Start with the simplest version of the puzzle: **AROP**, the EU's standard income-poverty rate, does not rank Greece as uniquely extreme — elevated, 4th-highest of 27 in the most recent year, but not standing alone at the top. Subjective poverty does: 1st of 27, by a wide margin, every year since 2011. This report treats that gap as a measurement-and-explanation problem, not as households being wrong.

The chart below shows AROP against subjective poverty for every EU country plus the EU average, in the most recent year available.

AROP vs. subjective poverty, EU countries, 2025

Sorted by subjective poverty.



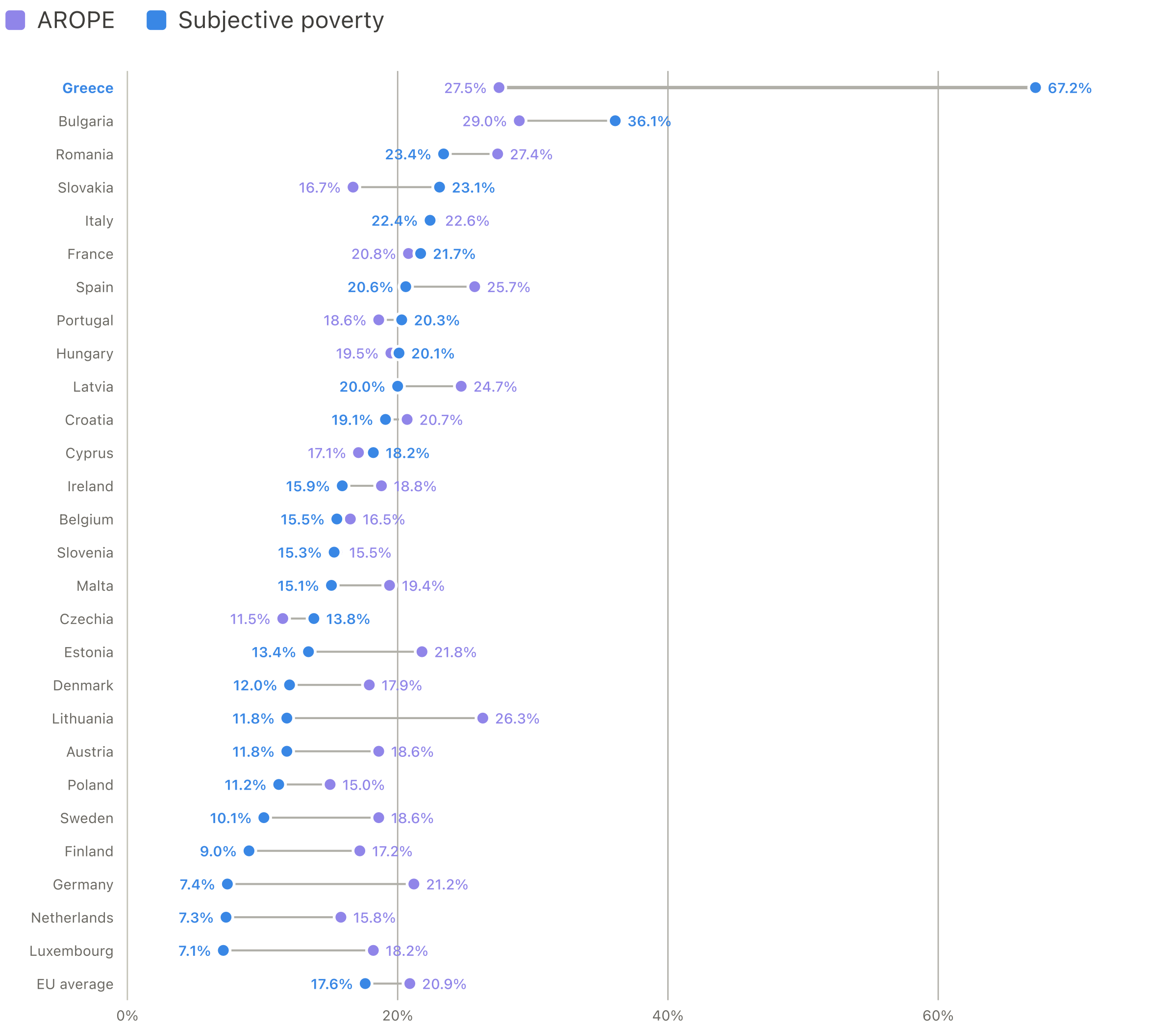
Source: Eurostat, ilc_li02 & ilc_sbjp01 2025

Core finding **Greece's AROP-vs-subjective gap is the largest in the EU by a wide margin, and not close.** Subjective poverty exceeds AROP by 47.6 points in Greece; the next-largest gap in that direction is Bulgaria's, at 14.9 points — Greece's gap is more than three times the runner-up's. Most EU countries sit far closer to agreement between the two measures, in either direction.

The EU already has an answer of sorts to "isn't AROP too narrow." **AROPE** — "at risk of poverty *or* social exclusion" — is Eurostat's own broader official measure, adding severe material deprivation and very-low household work intensity to income poverty. It is the official system's own recognition that income poverty alone doesn't capture everything. The question this report asks next is whether that broader measure is broad enough for Greece.

AROPE vs. subjective poverty, EU countries, 2025

Same countries, same sorting, the EU's broader official measure instead of AROP.



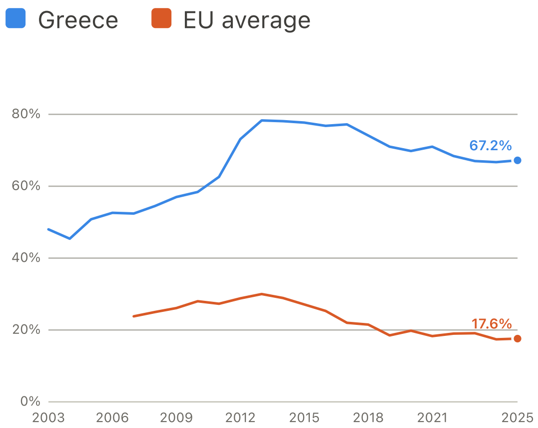
Source: Eurostat, ilc_peps01n & ilc_sbjp01 2025

AROPE narrows the gap, but does not close it. Subjective poverty exceeds AROPE by 39.7 points in Greece — smaller than the 47.6-point AROP gap, but still by far the largest gap of any EU country in either direction (next: Bulgaria, 7.1 points). AROPE matters here not as a replacement for AROP, but as the official system's own recognition that income poverty alone is too narrow — and even on its own terms, it doesn't fully answer the question it raises. That's the motivation for Movement 4: if the official broader measure accepts that hardship is multidimensional, this report tests those dimensions separately rather than treating AROPE as one fixed bundle (Movement 4 onward).

Greece's own headline numbers here (67.2% subjective poverty vs. 17.6% EU average; 19.6% AROP vs. 16.3% EU average; 27.5% AROPE vs. 21% EU average) are independently reported the same way by Greek data-journalism outlet Greece in Figures, drawing on the same primary Eurostat/ELSTAT data this report uses directly. That page covers Greece against the EU average only, not the full 27-country breakdown shown above — the country-by-country charts themselves are this project's own construction from the live Eurostat API, not reproduced from theirs.

Subjective poverty

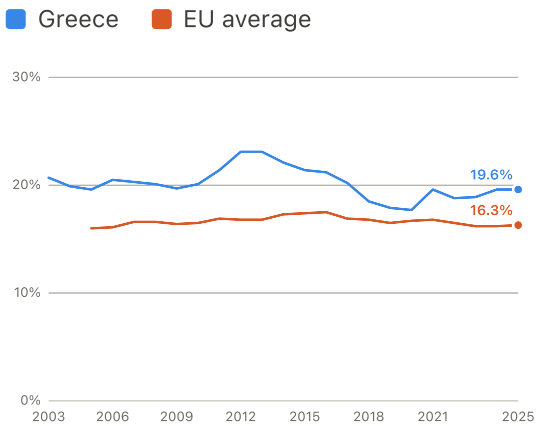
Share of households who say making ends meet is difficult or very difficult



ilc_mdes09 / ilc_sbjp01 2003–2025

AROP income-poverty rate

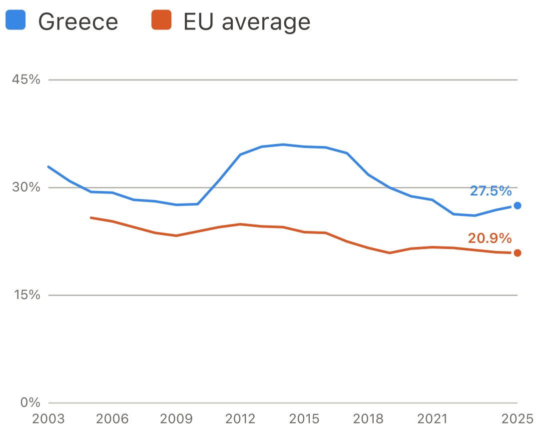
Income below 60% of the national median



ilc_li02 2003–2025

Broader official measure (AROPE)

Income poverty, severe deprivation, or very-low-work household



ilc_peps01 / ilc_peps01n

2003–2025, spliced definition

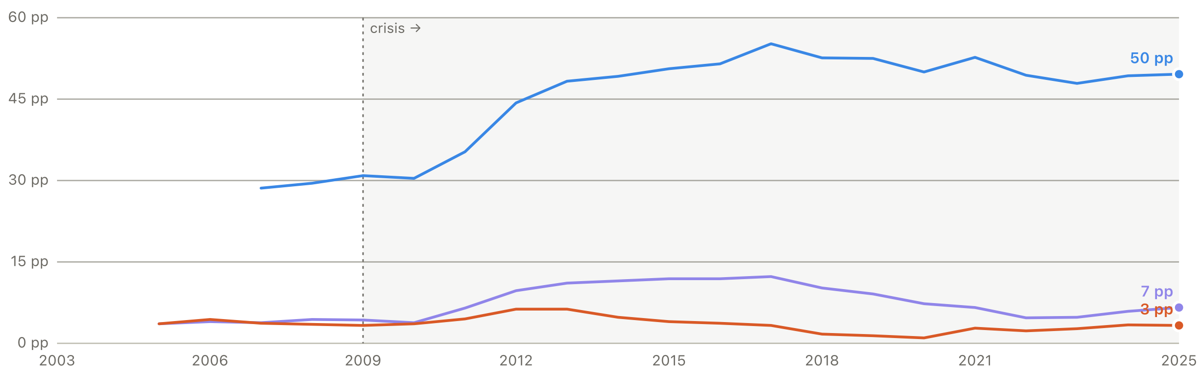
Already high before the crisis. In 2003, before any hint of the debt crisis, 48% of Greek households already said making ends meet was hard — more than double the AROP income-poverty rate of 20.7%. So this isn't just a crisis story layered on a healthy baseline; the gap was there from the start.

HOW BIG IS THE GAP, AND HOW UNUSUAL IS GREECE?

Two more views of the same divergence: the size of the gap between Greece and the EU average each year, and Greece's rank against other EU countries. Both official poverty concepts are shown here so the comparison does not silently switch measures: AROP is the income-poverty rate; AROPE is the broader poverty-or-social-exclusion rate.

Greece minus EU average, percentage points

Subjective poverty gap AROPE gap AROP gap

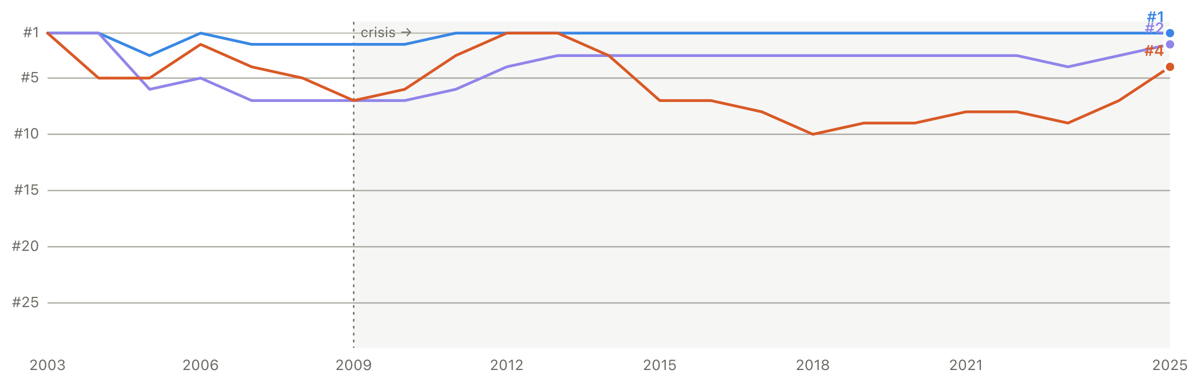


Positive = Greece worse than EU average

2005–2025 where EU aggregate coverage exists; AROPE added as broader official benchmark

Greece's EU rank, 1 = highest poverty rate

Subjective poverty rank AROPE rank AROP rank



Lower on chart = worse rank (closer to 1st)

EU member states only, membership as of each survey year

"Greece has one of the highest official poverty or exclusion rates in Europe" is well supported — AROP and AROPE both place Greece high in the EU distribution. But that's still a much less extreme statement than **"Greece has by far the highest subjective poverty rate in Europe,"** which is the accurate headline: 1st of 27–28 countries every year since 2011. The point is not that AROP and AROPE are useless; it is that neither official benchmark captures how unusually far Greek households' own assessment sits above the rest of Europe.

The disadvantage is also unusually broad, not confined to the headline measures. Across 25 indicators with an unambiguous worse-direction — excluding subjective poverty itself and every variable used in the models, so the measure cannot restate what this report sets out to explain — the share placing Greece in the EU's worst quintile rose from **21% before the crisis** (2005–2010) to **58% from 2012 onward**, reaching **68% in 2024**. Greece ranks 1st of 27 on this measure; Bulgaria, next highest, is at 50%. The step change dates to 2011 and has not reversed in the fourteen years since, and it does not depend on where the cutoff is drawn: Greece is 1st of 27 whether "worst" means the bottom 10%, 20%, 25% or a third of the Union.

What this is, and what it is not. This is descriptive corroboration — the fourth evidence tier — not a mechanism. The same measure was constructed as a model predictor and tested in a pre-registered family of six; it returned null (p=0.083, FDR 0.287) and is reported here only as a summary of Greece's condition, never as an explanation of it. Two constructions keep it honest: the worse-direction of every indicator is declared in advance rather than inferred, and the outcome plus all seven Model C-LTU covariates are removed before the share is computed. Full construction and the null result are in `scripts/50_persistence_share.py`. The statistical appendix lists all 25 indicators individually, showing where Greece sat in the EU distribution on each at its earliest usable year and where it sits now — so the composite can be audited rather than taken on trust.

EU aggregate for subjective poverty has no coverage before 2007; EU aggregate composition also changes over time (more countries joined, then the UK left) — see Methods for exact years. In 2003–2004 only 6–13 countries had comparable data yet, so Greece's early rank is indicative only, not the level.

MOVEMENT 2

The ruler moved

The question: AROP is defined against the current national median. When the median itself collapses, what happens to the measure — and what does a fixed yardstick show instead?

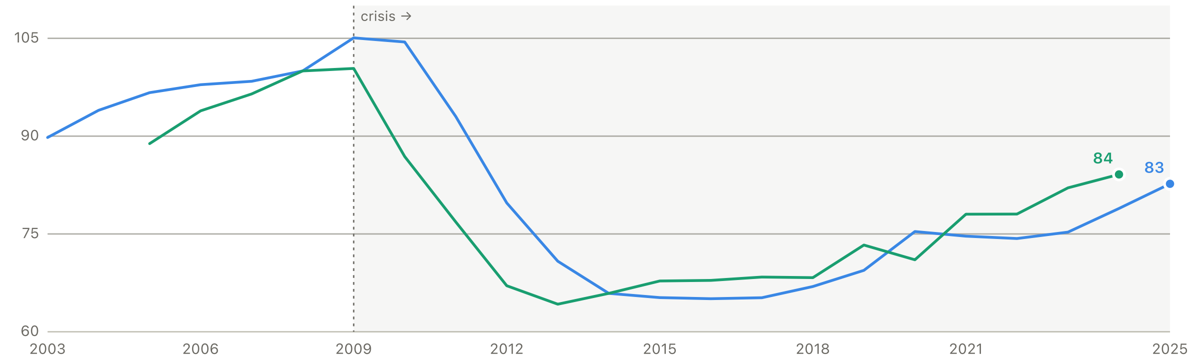
Why AROP behaves differently: the moving threshold

Here's the mechanical reason AROP stayed so calm through a historic income collapse. Its poverty line is defined as 60% of the national median income — not a fixed euro amount, but a moving target that resets every year. If everyone's income falls together, the line falls with it, and the share of people below the line barely changes — even though everyone underneath it is now poorer in real terms.

The poverty line itself, in real terms (2008 = 100)

Greece's actual euro poverty threshold, adjusted for inflation, vs. real household income

AROP threshold, real (idx 2008=100) | Real household income (idx 2008=100)



Threshold deflated using Greek inflation, chained from 2008 | ilc_li01, tepsr_wc310

The poverty line itself fell from 100 (2008) to a low of 65 in 2016 — a 35-point real decline — tracking real household income almost exactly (which fell to 68 the same year). Over that same stretch, the measured poverty *rate* barely moved, from 20.1% to 21.2%, because the ruler shrank along with the thing it was measuring.

Method: the poverty threshold Eurostat actually used each year (single-adult household, 60% of that year's median income, in euros) is deflated to constant 2008 prices using Greek consumer-price inflation. See Methods for the full definition.

3

A fixed yardstick: anchored poverty approximated

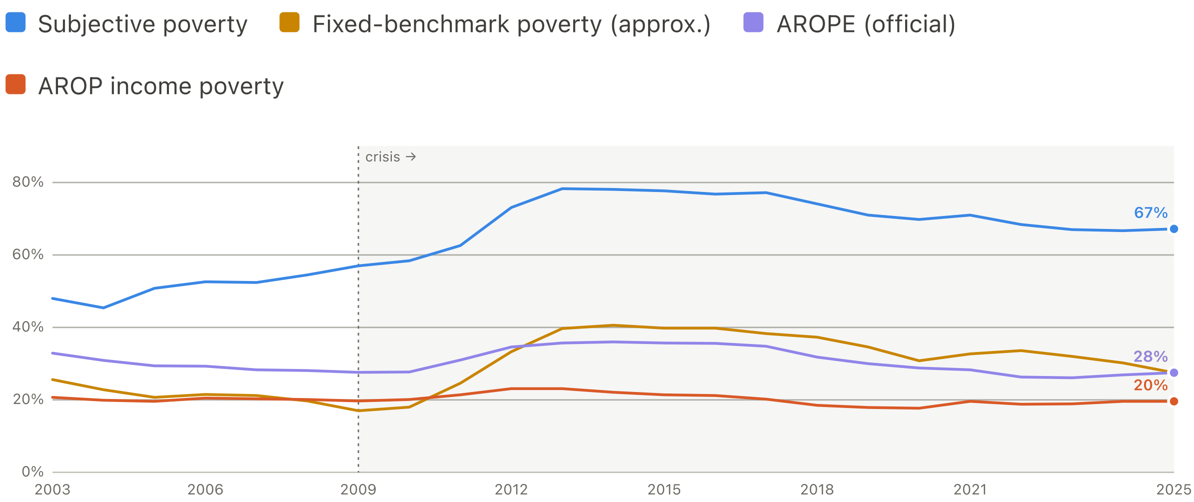
The chart above shows the ruler shrinking. This section asks the more direct question: what if we just didn't let the ruler shrink? What if "poor" still meant, ten years later, what it meant in 2008 — adjusted only for inflation, not for how far the whole country's income had fallen?

Eurostat doesn't publish that measure for Greece back to 2008 (their own version of it is anchored to 2019, which misses the crisis entirely). So this report builds an approximate version: each year, Eurostat publishes the poverty rate at four different income levels (40%, 50%, 60%, and 70% of that year's median). Fitting a smooth curve through those four points recreates a rough picture of the whole income distribution, accurate enough to read off "what share of people would fall below a 2008-level income" in any later year.

There's a fourth, genuinely official measure worth adding here, precisely because it needs no approximating: Eurostat's broader headline indicator, **AROPE** ("at risk of poverty *or* social exclusion"). Rather than income alone, AROPE counts anyone who is either income-poor, *or* severely materially and socially deprived, *or* living in a household with very low work intensity — whichever applies. Because two of those three conditions aren't tied to a shifting median income the way AROP is, AROPE doesn't have exactly the same moving-yardstick weakness — and it turns out to behave more like subjective poverty than plain AROP does, without any approximation involved.

Four measures of poverty, side by side

AROP vs. AROPE vs. fixed-benchmark poverty (approx., 2008 base) vs. subjective poverty — all shown as a percentage (population-weighted for the first three; see Methods for the subjective series' household-vs-population weighting)



Fixed-benchmark series is approximated; AROPE and AROP are official Eurostat figures — see Methods

ilc_li01, ilc_li02, ilc_mdes09, ilc_peps01/ilc_peps01n

Core finding **The fixed-benchmark measure and AROPE both behave far more like subjective poverty than the official AROP rate does — but for two different reasons, not the same one.** The fixed-benchmark line roughly doubled during the crisis — from about 20% (2008) to a peak near 41% (2014) — while AROP barely moved (20% → 22%) over the same span, because its *threshold* fell with the collapsing median (Movement 2). AROPE's good performance has a more mundane explanation: it's a union of AROP with severe material deprivation and low work intensity, and this project's own data already shows deprivation tracking subjective poverty closely on its own — AROPE mostly inherits that. It doesn't fix AROP's moving-threshold problem at all; the AROP-poor people inside AROPE are still counted against that same shifting line. What it adds is a different, complementary point: AROPE moved from 28% (2008) to a peak of 36% (2014) and tracks subjective poverty's year-over-year changes almost as closely as the fixed-benchmark line does (correlation 0.74 in year-over-year changes, 0.90 once each series' own trend is removed — both far above AROP's 0.60 and 0.67 on the same two tests), and it does that with no approximation at all, since Eurostat calculates it directly from survey data every year.

AROPE has its own methodology break, exactly like the material-deprivation series it partly draws on: figures through 2020 use the legacy definition, 2021 onward use a revised one (both from Eurostat, spliced here at the point where the two vintages meet rather than blended). Greece's AROPE rank among EU states has stayed in a narrower 2nd–7th band across the whole period — elevated, like AROP, but nowhere near subjective poverty's 1st-of-27 every year.

CHECKING THE APPROXIMATION AGAINST A REAL EUROSTAT NUMBER

A good curve fit through four points doesn't prove the method extrapolates accurately *beyond* those points — and the years that matter most here (2012–2014) require exactly that kind of extrapolation. So the same method was tested where it can actually be checked: Eurostat does publish an official fixed-benchmark poverty rate anchored to **2019** (rather than 2008). The same four-point method was used to estimate that 2019-anchored series independently, then compared year by year to Eurostat's real published numbers.

The method checks out reasonably well, with one honest caveat. Across 2019–2025, the approximation was off by 1.2 percentage points on average (worst case: 3.4 points, in 2022), and consistently tracked the real series (correlation 0.89). It also ran slightly high — a small, consistent overestimate of about 1 point. But none of the validated years required much extrapolation, because Greek incomes didn't move nearly as far from 2019 as they did from 2008 during the crisis. The 2012–2014 estimates above involve real extrapolation that this check couldn't directly test — so treat the crisis-era *shape* (a sharp rise, roughly doubling) as well supported, and the exact numbers in those specific years with a bit more caution than the validation alone would suggest.

Method: income is treated as approximately log-normally distributed; a two-parameter curve is fit to the four published (threshold, poverty-rate) points each year via a standard probit-linear regression (fit quality $R^2>0.99$ in every year). Full validation numbers (mean error, worst-case error, bias, correlation) are in Methods.

A SECOND, MORE RELEVANT CHECK: AGAINST ACTUAL CRISIS-YEAR MICRODATA

The check above tested the method in years that turned out not to need much extrapolation. A separate, independent study gives a better test — one that covers the crisis years directly. Andriopoulou, Kanavitsa & Tsakloglou computed Greece's anchored poverty rate for 2007–2016 using the actual underlying household survey data (not an approximation from four summary points, the real income of every household in the sample). Their number, for income year 2013: **48% of the population poor** by a 2007-anchored measure.

A year-labelling note before comparing: their "2013" refers to the income year. This report's series is labelled by EU-SILC *survey* year, which (as noted elsewhere in this report) generally reflects income from the *previous* calendar year. So their income-year-2013 figure lines up with this report's **survey-year 2014**, not 2013.

Independent microdata points in the same direction, and finds an even larger crisis-era effect. This report's survey-year-2014 estimate is 40.6%, about 7–8 points below Andriopoulou et al.'s 48%. That gap shouldn't be read as proof this report's method has a specific, quantifiable bias, though — the two exercises differ in more than just the approximation (household microdata vs. a four-point aggregate reconstruction, the exact price index and anchor-year definition used, and data vintage all differ too), so the difference can't cleanly be attributed to any one of them. What it does provide is real external corroboration of the qualitative story: an independent, microdata-based study finds a fixed pre-crisis poverty line reveals far more deterioration than contemporaneous AROP shows — if anything, more than this report's own reconstruction shows.

Method & a caveat on comparability: Andriopoulou et al. anchor to 2007 rather than 2008 and use Greece's national SILC microdata directly rather than the four-point Eurostat summary this report relies on. Andriopoulou, E., Kanavitsa, E., & Tsakloglou, P. Decomposing poverty in hard times: Greece 2007–2016. *Revue d'économie du développement* (2019); also circulated as LSE Hellenic Observatory GreeSE Paper No. 149 (2020), the version consulted for this report.

4

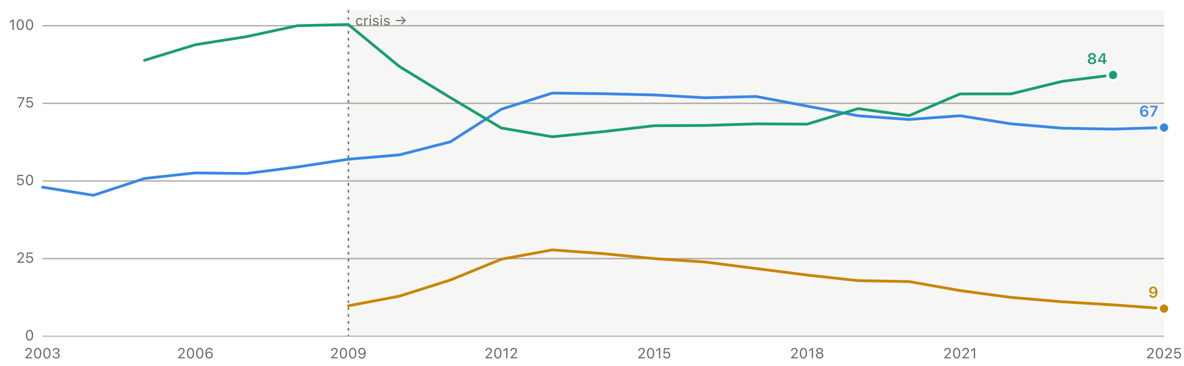
Does subjective poverty track real economic conditions in Greece?

If subjective poverty were mostly a matter of mood or expectation, it would drift loosely from the actual economy. In Greece, it moves closely with unemployment and household income — and that relationship survives a real attempt to break it.

Subjective poverty, unemployment, and real income — indexed together

All three lines share one scale to make the turning points easy to compare — they're different units (share of households, % of the labour force, and an income index), so only the shape should be compared, not the exact levels

Subjective poverty (%) Unemployment rate (%) Real income (idx 2008=100)



Real income idx 2008=100 · Unemployment %, age 15–74 2009–2025 (unemployment data starts in 2009)

All three turn in the same year. Subjective poverty peaked in **2013 (78.3%)** — the exact same year unemployment peaked (27.8%) and real household income hit bottom. It then fell through the 2013–2019 recovery as unemployment nearly halved, held flat through COVID, eased further through 2023–2024 as unemployment kept falling, and ticked back up slightly in 2025.

A STRICTER TEST: DOES IT SURVIVE REMOVING THE SHARED TREND?

Two things that are both rising or both falling for years at a time will often look strongly correlated even if they aren't really connected week to week — they just share a slow-moving trend. To rule that out, each relationship below is tested three ways: the raw correlation, the correlation of *year-over-year changes* (does a good year for one line up with a good year for the other?), and the correlation after removing each series' own long-run trend. If a relationship survives the second and third test, it's much more likely to be a real, short-run connection rather than two lines that just happen to share a shape.

VARIABLE	RAW CORRELATION	YEAR-OVER-YEAR CHANGE	TREND REMOVED
Long-term unemployment rate (12mo+)	0.93	0.92	0.86
Real household disposable income	-0.93	-0.53	-0.91
Severe material & social deprivation, new	0.92	0.38 (n.s.)	0.11 (n.s.)
AROP threshold, real terms	-0.88	-0.65	-0.78
Housing cost overburden	0.86	0.51	0.86
Severe material deprivation, legacy	0.82	0.65	0.62
Real GDP per capita	-0.82	-0.49	-0.82
Households in arrears	0.82	0.47	0.67
Unemployment rate	0.81	0.86	0.80
Cannot keep home adequately warm	0.80	0.63	0.89
Real wages, compensation per employee	-0.79	-0.21 (n.s.)	-0.42 (n.s.)
Cannot face unexpected expense	0.76	0.52	0.58
Youth unemployment rate (15-24)	0.71	0.79	0.81
Real household consumption per capita	-0.70	-0.55	-0.84
Employment rate 20-64	-0.69	-0.72	-0.77
HICP inflation, headline	-0.54	-0.25 (n.s.)	-0.67
AROPE (poverty or social exclusion)	0.48	0.74	0.90
AROP	0.19	0.60	0.67
Income inequality, S80/S20 ratio	0.15	0.64	0.82

n.s. = does not survive Benjamini-Hochberg FDR correction (applied separately within each column, across all 19 variables tested together, at $\alpha=0.05$) — the right threshold to use with this many comparisons run at once, not the uncorrected $p<0.05$ a single test would use.

Most relationships hold up under correction for testing all nineteen at once — a couple don't, and AROP turns out more interesting than it first looked. Sixteen of nineteen variables keep a real relationship on year-over-year change after FDR correction; seventeen of nineteen keep one after removing each series' own long-run trend. Unemployment, income, jobs, housing costs, missed bill payments, and one deprivation measure are all among the survivors — some (unemployment, housing costs) even get stronger under the stricter tests. **The strongest of all, by a clear margin, is long-term unemployment** (the share out of work 12 months or more): raw correlation 0.93, year-over-year 0.92, trend-removed 0.86 — every other variable in this table weakens more once the shared trend is removed. Movement 3 goes further with it, testing it directly in the cross-country model. **The EU's own broader poverty-or-exclusion measure, AROPE, is in this table for the first time here and survives comfortably on both stricter tests** (year-over-year 0.74, trend-removed 0.90) — despite a weak raw/level correlation (0.48) that would have looked unimpressive on its own. **Two variables fail decisively, on both stricter tests:** severe material & social deprivation — the newer, revised 13-item EU-SILC measure (distinct from the legacy 9-item version above, which does survive) — looked striking on the surface (raw correlation 0.92) but drops to essentially nothing (0.11) once the shared trend is removed, and it's only usable from around 2015, so this test has just 10–11 years of data to work with, too short to tell a real relationship from coincidence. Real wages (compensation per employee) also fails both: its year-over-year change isn't reliably related at all, and its trend-removed relationship, which looked borderline before correction, doesn't survive once the other eighteen tests run alongside it are accounted for. **One variable is a genuine mixed case, reported as such rather than smoothed over:** headline HICP inflation fails the year-over-year test but survives the trend-removed one. **And here's the interesting nuance on AROP:** its raw correlation is weak (0.19, and doesn't survive correction on its own), but its year-over-year change correlation is a respectable 0.60 and does survive. In other words, AROP isn't a bad indicator — its problem is specifically about the *level* it reports during a society-wide income collapse (Movement 2), not about whether it responds to real change at all. When the economy gets worse or better from one year to the next, the AROP income-poverty rate does move in the right direction; it just can't tell you how bad things have gotten overall, because its own yardstick moves too.

Method & a further check: a similar test was also run looking one and two years forward and back (not just the same year), on year-over-year changes rather than levels, to see whether the timing of the relationship makes sense without reintroducing the shared-trend problem the year-over-year test above already guards against. For unemployment, the clearest result there, the same-year and prior-year relationships are essentially tied (both around 0.86) and clearly the strongest points in the window, tapering off further out in either direction. This is offered as a timing clue, not as further proof against a coincidental relationship — that proof is what the year-over-year and trend-removed columns above already provide.

Does the relationship hold across Europe, or is it just Greece?

Everything so far either looks at all EU countries pooled together, or at Greece by itself. Those are different questions. Does a country's *own* subjective poverty move with its *own* changing conditions — a pattern that should show up all across Europe, not just in Greece? To test that, this compares changes *within* each country over time, rather than comparing countries' overall levels against each other (that second, level-based comparison is the subject of Movement 4, and needs its own separate model — this one, by design, can't see Greece's overall level at all, only its year-to-year wiggle).

VARIABLE	COMPARING COUNTRIES' LEVELS	WITHIN EACH COUNTRY OVER TIME
Unemployment rate	2.21	1.07
Income (purchasing-power adjusted)	-0.88	-0.87
Material & social hardship	1.15	1.17
AROP (income poverty rate)	-0.61	0.18

27 EU states, 2015–2024. Both columns control for unemployment, income (purchasing-power adjusted), material hardship, and the AROP income-poverty rate together, plus year-to-year EU-wide shocks; the second column also removes each country's own fixed baseline level.

Method: a two-way fixed-effects panel regression (country and year fixed effects, country-clustered standard errors) against a pooled regression with year effects only.

There is clear evidence the relationship generalizes beyond Greece — though not equally strongly for every variable. Material hardship is robustly linked to subjective poverty within countries over time (a strong, reliable result). Unemployment points the same direction but is right at the edge of conventional statistical reliability. AROP, once again, comes back weak and unreliable in both comparisons — and even flips sign between them — reinforcing that income poverty alone is a poor stand-in for lived financial strain regardless of how the question is framed.

ANSWER TO QUESTION 1

Yes, in large part. Reported difficulty making ends meet rose and fell in step with real income, unemployment, and material hardship — a relationship that survives the stricter year-over-year and trend-removed tests, not just a raw correlation, and holds up as a general European pattern, not something unique to Greece. Much of why AROP looked so much calmer is arithmetic: its threshold is pegged to the national median income, and the national median collapsed along with everyone's income. A fixed, pre-crisis benchmark instead shows poverty roughly doubling — closely matching what households were actually reporting, and independent microdata research finds an even larger effect (see Movement 2) — though the exact estimate in the deepest crisis years carries more uncertainty than the rest of the series.

MOVEMENT 3

How broad the deterioration was

The question: A single headline number says how far Greece fell on one measure. It does not say how much of Greek life the fall touched. Before any model, this movement reports the record in the units people actually live in — wages, prices against wages, housing, work, migration, age.

11

The scars beneath the gap

This section has one central test and one corroborating record. The central test asks whether **duration** matters: long-term unemployment first, then sustained exposure under full nested validation. The corroborating record asks what that history looked like in output, pay, prices, housing, migration, age, expectations, and work. Those findings strengthen interpretation but are not eight equal mechanisms or eight scorecard additions.

1. Duration
11.6 → 3.9

2. Sustained exposure
legacy nested residual +2.70; rank 19/27

3. Material record
GDP, pay, prices, housing

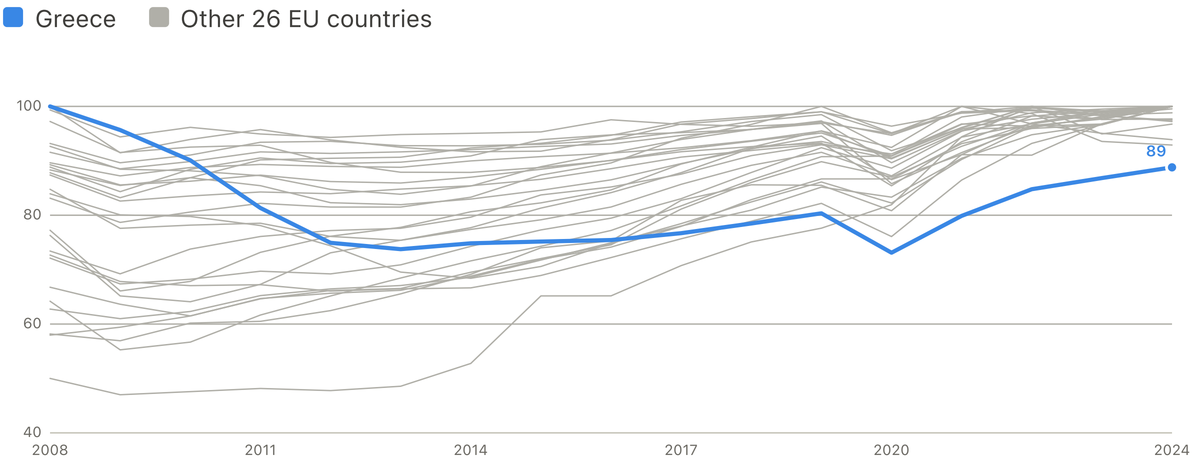
4. Social record
migration, age, expectations, work

THE MATERIAL RECORD: STILL BELOW ITS OWN PRE-CRISIS PEAK

The typical EU country that suffered a real crisis-era downturn recovered from it in about **3 years**. Greece has not recovered in **16**.

Real GDP per capita, indexed to each country's own historical peak (2008–2024)

Every EU country's line is shown; Greece's unresolved shortfall is the deepest and longest



Source: Eurostat, sdg_08_10 2008–2024

Core finding **Almost every EU country that had a real crisis-era downturn recovered within a few years; Greece's unresolved loss is exceptional in both scale and duration.** The debt crisis drove a 26.2% decline from the 2008 peak to a first trough in **2013**. Recovery followed, but COVID pushed the index to a marginally lower trough in **2020**. By 2024 Greece was still 11.2% below 2008. The median EU recovery time was 3 years. Finland also never fully regained its 2008 peak, but is only 2.3% below it; Luxembourg is 6.1% below a much more recent 2021 peak. Those cases prevent an "only country" claim, while leaving Greece's sixteen-year shortfall clearly the deepest and longest unresolved pre-crisis loss.

Methods, robustness and caveats

Not just a bigger recession — a structurally different crisis. The academic and EU-institutional literature on the Greek crisis is consistent on this point: Greece combined a sovereign-debt crisis with an external-competitiveness crisis, a sudden stop in capital inflows, weak tax and fiscal institutions, and euro-area constraints (no currency to devalue, no early crisis-management tools) that most EU countries never faced at the same time. Gourinchas, Philippon & Vayanos (2016) call it "one of the worst [crises] in history, even in the context of recorded 'trifecta' crises" — sudden stop, sovereign debt, and lending-boom-bust together — and find Greece's output collapse was "significantly more severe and protracted" than any comparable episode since 1980, with real income per capita falling every single year from 2007 to 2013 (a cumulative 26% drop) and barely rising since. The European Stability Mechanism's own account adds the domestic side: cheap euro-era borrowing let Greece delay necessary reforms through the 2000s, while weak tax administration and a 2009 revelation of misreported budget data eroded investor confidence and triggered the loss of market access. That matters here because it means the depth and duration of Greece's scarring isn't a story about Greek households reacting unusually to an ordinary downturn — the downturn itself was unusual. Full citations in Methods.

How far below its own historical peak, real GDP per capita, 2024

COUNTRY	% BELOW ITS OWN 2008–2024 PEAK
Greece	11.2%
Estonia	7.1%
Luxembourg	6.1%
Ireland	3.3%
Austria	2.8%
All other EU countries (18 of 27)	0.0–2.5%, most fully recovered

"All-time high" runs from 2008 (before the earliest usable point for most countries) through 2024, so a country that dipped in 2009 or 2020 but has since grown past its old peak counts as fully recovered (0%). This is the current-snapshot view; the finding above and the chart use the fuller trajectory, including when each country's worst dip happened and whether it has ever recovered from it.

Methods, robustness and caveats

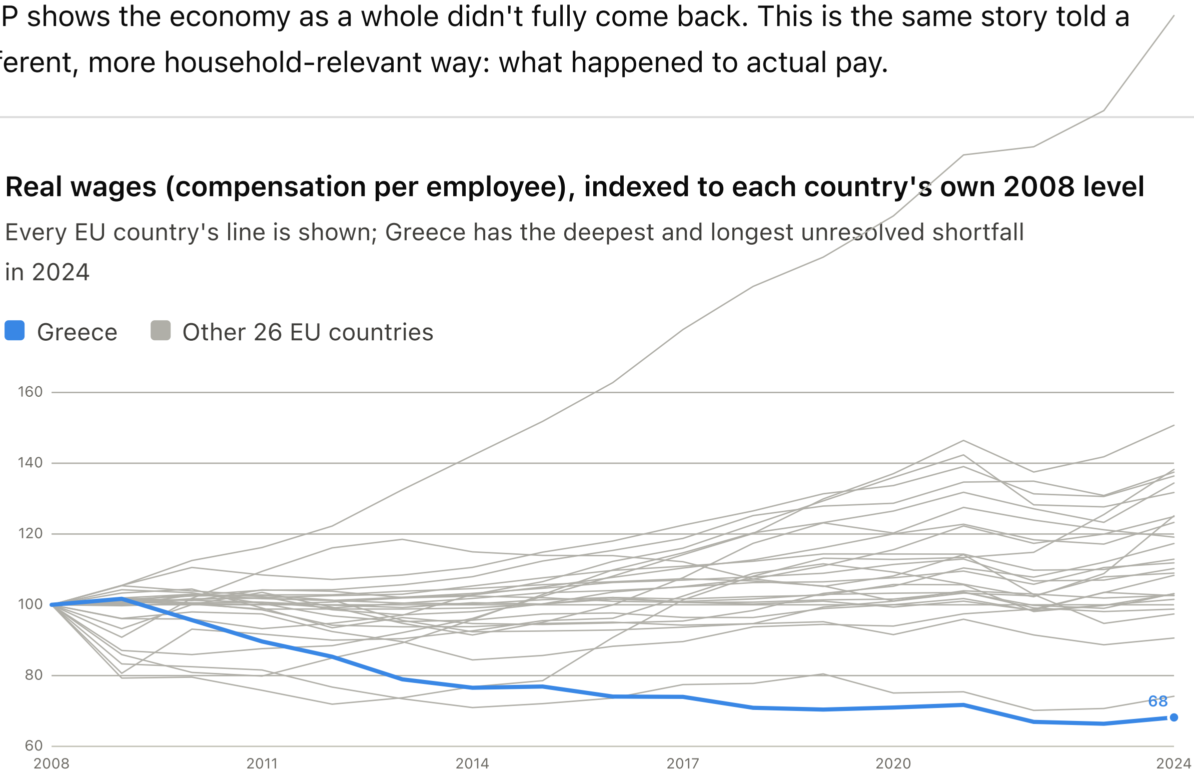
Method: built from Eurostat's real GDP per capita series (`sdg_08_10`), 2008–2024. Two related but distinct measures, kept separate: (1) each country's running historical maximum minus its current value, as a percentage — the "how far below peak right now" snapshot in the table above, added directly into the panel regression as a year-by-year predictor (statistically significant, p=0.045, and it cuts Greece's out-of-sample gap (the legacy section) from 11.6 points to 7.1, moving Greece from clearly the largest gap in the EU to 4th-largest of 27, behind Luxembourg, Cyprus, and Portugal); (2) each country's worst historical drawdown and whether/when it recovered from it — the "years to recovery" finding above, found using the same maximum-drawdown method as an earlier exploratory script in this project, but now checking explicitly whether and when each country's value returned to the peak that preceded its worst dip. These can disagree (a country can have recovered from its worst crisis years ago and still be marginally below a very recent, minor peak) — Luxembourg is exactly this case. Consistent with published research on the underlying mechanism: Baldini, Gallo & Torricelli (2020) find that a past income-scarcity spell continues to depress subjective make-ends-meet assessments up to two years later across Europe, even controlling for current income (full citation in Methods). **Update, tested directly below:** once long-term unemployment is added to the same regression alongside this scarring-stock variable, the scarring stock's own contribution stops being statistically significant (p=0.61) and the model's fit doesn't improve beyond what long-term unemployment alone already achieves. Read together, this GDP-level shortfall is best understood as useful macro context — the economy as a whole didn't come back — while long-term unemployment, covered later in this section, looks like the sharper, more household-facing version of the same scarring story.

THE PAYCHECK DIDN'T RECOVER EITHER

GDP shows the economy as a whole didn't fully come back. This is the same story told a different, more household-relevant way: what happened to actual pay.

Real wages (compensation per employee), indexed to each country's own 2008 level

Every EU country's line is shown; Greece has the deepest and longest unresolved shortfall in 2024



Source: Eurostat, nama_10_lp_ulc (deflated by prc_hicp_aind)

2008–2024

Chart is capped at 160 (2008=100) to keep the meaningful Greece-vs-EU comparison zone readable; a handful of Eastern European countries with much lower 2008 wage levels grew well past that line (Bulgaria to 267, Lithuania to 151) and run off the top of the frame — a different, catch-up-growth story from Greece's decline, not comparable on this chart.

Descriptive check **By 2024, Greek real wages were still 31.8% below their 2008 level — the largest shortfall of any EU country.** Hungary was second at 25.8% below and Italy third at 9.4% below, while most countries had recovered or grown substantially (Bulgaria +166%, Lithuania +51%, Romania +38%). Greek real wages peaked in 2009 and reached their low point in 2023. This is paycheck-level evidence of prolonged loss, but it does not independently improve the preferred model.

Methods, robustness and caveats

How this relates to real wages' correlation with subjective poverty, and to the existing models: at the level, real wages track subjective poverty about as strongly as any variable in this report (r=−0.79, p<0.0001) — but on year-over-year changes, that relationship is weak and not statistically significant (r=−0.21, p=0.35; detrended, r=−0.42, p=0.049 — borderline before correction, and it does not survive FDR correction for the nineteen variables tested together in Movement 2's table, adjusted p=0.052). The same pattern this report has flagged for other variables: a level correlation that likely reflects a shared crisis-era trend, not tight year-to-year co-movement. Real wages do not move tightly with subjective poverty year to year, but their long-term collapse helps explain why Greece's hardship reads as structural rather than cyclical. Checked before adding it anywhere near the cross-country regression: real wages correlate only modestly with the income and scarring variables already in the model (PPS-adjusted income r=–0.12, real GDP per capita r=–0.12, the scarring stock r=–0.33) — not redundant with them. But adding it to the richest existing cross-country model (the legacy section's Model C) barely moves anything: R^2 0.892→0.893, the coefficient isn't statistically significant (p=0.33), and Greece's out-of-sample gap only inches down (11.6→11.2, rank unchanged at 1st of 27). Not added to the scorecard as a result — it isn't redundant with the existing variables, but it also doesn't independently explain more of the cross-country gap once they're already accounted for. Kept here as a strong descriptive finding about Greece's own paycheck recovery, not a model addition. **Evidence role:** material-scarring context, not an independent scorecard variable — it thickens the case that the scarring is real and paycheck-level, distinct from its statistical strength.

A SMALLER PAYCHECK FACING ORDINARY PRICES

A units note before the next two numbers: the 31.8% figure above is *real* wages — inflation-adjusted purchasing power, tracked against Greece's own 2008 level. What follows is a different, *nominal* comparison — the actual euro paycheck, compared against actual euro prices, benchmarked against the rest of the EU rather than against Greece's own past. Both are correct; they're just answering different questions.

Greece is not expensive in the way Denmark or Ireland is expensive. Its overall consumer price level is below the EU average. But Greek wages are much further below the EU average. Once prices are scaled by what a Greek paycheck actually buys, the picture reverses completely.

Greece, 2024: raw price level vs. wage-adjusted price pressure, by category

Price level and wage level are both indexed to the EU average (=100). Wage-adjusted pressure = price level ÷ wage level × 100 — above 100 means this category costs proportionally more of a Greek paycheck than of an EU-average paycheck.

CATEGORY	RAW PRICE LEVEL (EU=100)	EU RANK, PRICIEST	WAGE-ADJUSTED PRESSURE	EU RANK, MOST PRESSURE
Overall household consumption	**83.6**	**18th of 27**	**173.1**	**1st of 27**
Food & non-alcoholic beverages	106.2	9th of 27	219.9	2nd of 27
Housing, water, electricity, gas & fuels	73.5	20th of 27	152.2	2nd of 27
Transport	89.7	13th of 27	185.7	1st of 27
Restaurants & accommodation	86.6	20th of 27	179.3	1st of 27
Information & communication	123.2	6th of 27	255.1	1st of 27

Wage level: nominal compensation per employee, Greece as a share of the EU27 average (48.3% in 2024, 4th-lowest of 27). Source: Eurostat `prc_ppp_ind_1` (price levels) and `nama_10_lp_ulc` (wages).

Core finding **On overall consumption, Greece ranks 18th-most-expensive of 27 EU countries by raw price level — but 1st for wage-adjusted price pressure.** The reversal holds in every category tested, not just the overall basket: Greece ranks 1st or 2nd of 27 on wage-adjusted pressure for food, housing, transport, restaurants, and communication alike, even in the categories (housing, transport, restaurants) where its raw prices sit well below the EU average. Greek wages are so far below the EU average (48.3%, 4th-lowest) that even moderately-priced goods and services become disproportionately heavy on a Greek paycheck.

Over time, the gap has widened from both directions at once. Since 2008, Greek nominal wages are still 13.1% below their 2008 level. Over the same span, nominal prices rose: food up 41.6%, housing and energy up 33.6%, and the general price level up 27.5%. The paycheck shrank in absolute euro terms while the receipt grew.

Methods, robustness and caveats

What this is, and isn't. This is a comparative price-level index (Eurostat's harmonized category comparisons), not an actual household-budget survey or real supermarket receipts — it measures relative price levels across countries, not what any specific Greek household spends. Descriptive evidence, not a model finding: the granular category price data (food, housing, transport, restaurants, communication) has full 27-country coverage for only three years, 2022–2024 — too little time variation to support the year-fixed-effects panel regression used elsewhere in this report. Only the broader "overall consumption" category has a long enough series (2000–2024) to check against subjective poverty at all: level correlation r=0.70 (p=0.0003), but not on year-to-year changes (r=–0.06, not significant) — the same pattern this report has repeatedly flagged for other variables, most recently real wages just above. **Not added to the scorecard as a result** — kept here as strong descriptive evidence for why the gap feels structural, not as a tested cross-country predictor. **Evidence role:** material-scarring context — the same role real wages play just above, from a different angle (what a paycheck buys, not just its size).

OWNING YOUR HOME DOES NOT FULLY PROTECT YOU HERE

In most EU countries, owning a home outright is a powerful shield against housing-cost pressure: pay off the mortgage, and housing costs mostly disappear as a source of strain. Greece is different. Its housing problem doesn't stop at the renter-owner line.

Housing cost overburden by tenure status, 2024

GROUP	OVERBURDEN RATE
Mortgage-free owners — Greece	25.7%
Mortgage-free owners — next-highest in the EU (Sweden)	14.0%
Mortgage-free owners — typical EU range (most countries)	1–7%
Market-rate renters — Greece	37.4%

"Overburden" = housing costs exceeding 40% of disposable income. Source: Eurostat `ilc_lvho07c`.

Descriptive check **A quarter of Greek mortgage-free owners are still overburdened by housing costs — the highest share in the EU, and almost twice the next-highest country.** Most EU countries show a sharp divide: renters squeezed, outright owners largely shielded. Greece shows something different — renters are still worse off (37.4% overburdened) than owners (25.7%), but the gap between them is one of the EU's narrowest (Greece ranks 22nd of 27 on the size of that gap) precisely because owners are unusually burdened too, not because renters are spared. Greece's housing problem is not only a renter problem. It reaches households who, on paper, should be protected — consistent with (though this report does not directly test) pressure from property tax, energy costs, and maintenance on an older housing stock relative to income, rather than rent or mortgage payments themselves.

On ownership structure itself, Greece isn't unusual in how many people own homes (69–70% recently, close to the EU average) — it's unusual in how they own them: only 8–17% of Greek ownership is mortgage-financed, versus a stable 24–26% EU-wide, with the rest owned outright. That's plausibly consistent with inherited or self-built housing and an older housing stock rather than credit-financed purchase, though this report doesn't test the reason directly. One further, still-unfolding shift worth flagging: Greece's ownership rate has fallen faster than the EU average recently — 73.9% to 69.7% from 2020 to 2024, more than double the EU's own decline over the same years.

Methods, robustness and caveats

Correlation, labeled descriptive rather than causal: mortgage-free-owner overburden tracks Greece's own subjective poverty closely over time (r=0.90, 18 years) and correlates across all 27 EU countries in a single cross-section too (r=0.66, 2024) — both real, both worth reporting, neither treated here as evidence of a causal channel running specifically through ownership status. **Model role:** not a scorecard candidate. This series correlates at r≈0.9 with the `housing_cost_overburden` variable already in Model C-LTU (unsurprising, since it's a subcomponent of that total), and adding it on top changes little (R^2 0.914→0.917, Greece's out-of-sample gap moves from 3.9 to 5.1) with a coefficient that isn't statistically significant (p=0.14). The housing-cost channel is already captured in the scorecard; this section adds texture to *why* it's so severe in Greece specifically, not new independent explanatory power.

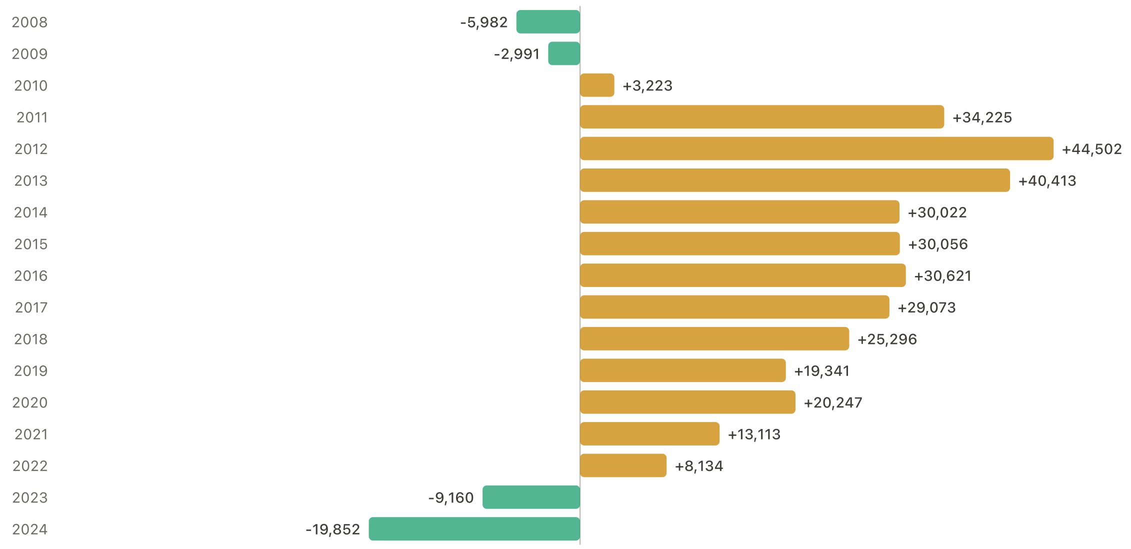
THE SOCIAL RECORD: THE CRISIS ALSO BECAME AN EXIT ROUTE

Income and jobs aren't the only things a household can lose during a sixteen-year, unrecovered crisis — people themselves can leave. This isn't tested as a variable in any regression here; it's a structural consequence of the same crisis already documented above, worth understanding in its own right.

Net migration of Greek nationals, 2008–2024

Positive = more Greek citizens left than returned that year. Negative = more returned than left.

Net migration of Greek nationals



Source: Eurostat, migr_emi1ctz & migr_imm1ctz, citizen = national 2008–2024

A large, crisis-linked outflow of Greek nationals, only recently starting to reverse. Net outflow rose from under 6,000 a year in 2008–2009 to a peak of 44,502 in 2012 — the depth of the crisis — and stayed elevated through the late 2010s. Cumulative net loss of Greek nationals, 2008–2024: **290,281 people, 2.6% of Greece's 2008 population.** The migration data on its own is not primarily a hardship measure the way housing costs or arrears are — it's evidence that during exactly the years income, jobs, and expectations were collapsing, a meaningful share of households treated leaving the country as the answer. Genuinely new and worth reporting plainly: 2023 and 2024 both show a *net inflow* — more Greek nationals returned than left, for the first time since before the crisis. The recent return is real. It does not erase the cumulative loss, or the years of disrupted careers and family separation behind it.

Methods, robustness and caveats

A specific push factor worth naming: youth unemployment. Youth unemployment (ages 15–24, checked directly as a candidate variable — Methods) peaked at 59.2% in 2013, one year after net departures of Greek nationals peaked in 2012. That timing is consistent with youth labor-market collapse as a push factor for emigration, but this report does not treat it as a causal migration model — it's a descriptive, same-era coincidence, not a tested relationship. Youth unemployment tracks subjective poverty strongly on its own (r=0.71 level, r=0.79 first-difference, r=0.81 detrended, Movement 2), but overlaps so heavily with long-term unemployment (above) that it adds no independent explanatory power once long-term unemployment is already in the cross-country model — kept here as descriptive context, not as its own model finding. **Evidence role:** social/demographic scarring context, alongside migration itself — not a scorecard variable, not a substitute for the duration mechanism above.

Methods, robustness and caveats

One EU country's emigration isn't automatically more severe than another's — check before claiming it. Ranked by cumulative net emigration of nationals as a share of population (25 EU countries with adequate 2008–2024 coverage), Greece is **5th of 25** at 2.7% — behind Lithuania (8.3%), Croatia (5.9%), Romania (4.3%), and Luxembourg (a different kind of case: an economic migration hub, not a crisis-emigration story). Several Eastern and Baltic countries lost a proportionally larger share of their population to emigration than Greece did. That's a real check against overstating Greece's case on this one dimension — the argument here isn't that Greece's emigration was uniquely severe in the EU, only that it was large, crisis-linked, and sustained for over a decade.

Methods, robustness and caveats

Why this looks different from other recent reporting on "brain drain reversal," and why that's not a contradiction: this report's own figures are annual *flow* data — how many Greek citizens left and returned in each calendar year, from Eurostat's migration-flow tables. A separate Greekonomics.gr article on brain drain specifically (distinct from the Mantés & Marinakis report cited elsewhere in this section; full citation in Methods) uses *stock* data instead — the total number of Greek-born working-age people resident abroad — and found that figure roughly flat from 2019 to 2022 before a modest rise by 2025, concluding that "brain drain reversal" claims based on that measure were premature. Both can be true at once: annual returns can now exceed annual departures while the accumulated stock of people who left during the crisis has only partly come home. A 2026 OECD review of Greek emigrants (with Greece's National Documentation Centre, EKT) puts a number on the flow finding directly — 46,000 Greek citizens returned in 2023 against 37,000 who left, the first positive year since the crisis began — consistent with this report's own Eurostat-derived figures. That same OECD review, using Greek census data (a different source again from either the flow or stock figures above), finds recent returnees skew young and highly educated: 54% of 2016–2021 returnees were aged 20–39, and three in five held a tertiary degree, versus 23% among Greek residents who never emigrated. This report's own migration-flow table cannot isolate the age or education profile of Greek nationals leaving or returning — that finding comes from the OECD/census source specifically, not from the Eurostat flow data charted above, and the two are kept separate here rather than conflated. Full citations in Methods.

Methods, robustness and caveats

"Brain drain" is a real but incomplete frame. Skilled emigration from Greece did not start with the 2010s crisis. Labrianidis & Vogiatzis (2013) argue the relationship runs in both directions: Greece's pre-crisis underutilization of its own highly educated workforce was itself one contributing factor in the debt crisis, which then further accelerated the outmigration of graduates — a mutually reinforcing dynamic rather than a crisis simply causing a one-way exodus. Pratsinakis (2022), in a dedicated academic study of the post-2010 emigration wave (34 in-depth interviews plus a 996-respondent survey in London and Amsterdam), makes the point more directly: reducing this migration wave to "brain drain" alone "reproduces a statist and economistic conceptualisation of migration far removed from the migrants' subjectivities," obscuring both the emigration of less-educated Greeks and the fact that most post-2010 emigrants describe a mix of motives, not pure economic compulsion. In his survey, only 27% of post-2010 emigrants said their decision was "enforced by circumstances in Greece" — 43% said emigrating was something they had "basically been wishing for" regardless, with the crisis giving them the final push. That doesn't weaken the finding above; a genuine, large, crisis-linked outflow, documented earlier in this section, coexisted with more varied individual motives than "forced exodus" alone implies. A separate, complementary data point: unemployment among highly educated Greeks rose to almost four times the EU-28 average during the crisis (Labrianidis & Pratsinakis, 2016, cited in Pratsinakis 2022) — and in a nationwide 1,237-household survey, two in three people who emigrated during the crisis years held a university degree.

Methods, robustness and caveats

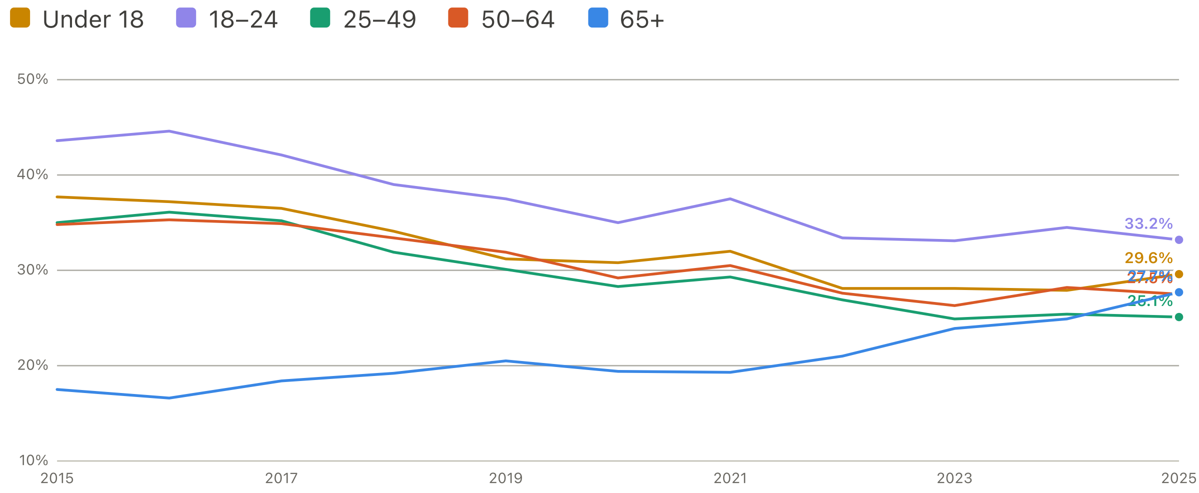
A third source, a similar but not identical shape: Pratsinakis (2022), using Eurostat data through 2019, also shows Greek citizens' emigration surging after 2010 and peaking around 2012, then "rather stabilised" at roughly 50,000 a year through the late 2010s rather than declining steadily — the same peak-then-plateau shape the OECD review describes (Methods, above), and not quite identical to this report's own Eurostat extraction, which shows more of a steady decline after 2012. Three legitimate sources, three slightly different shapes from the same broad Eurostat data family — consistent on the crisis-era surge and the 2012 peak, less so on exactly how quickly the outflow eased afterward. Reported honestly rather than smoothed into false agreement.

THE NATIONAL AVERAGE HID A GENERATIONAL REVERSAL

Everything above treats Greece as one population. It isn't. AROPE, AROP, and material deprivation are all published broken down by age — and the age breakdown tells a sharper, more specific story than "the aggregate got a little worse in 2025." This section is a distributional finding about who inside Greece carries the gap, not another candidate explanation for the cross-country residual above — it doesn't enter any regression in this report and should be read alongside the AROPE bridge (Movement 1) and the discussion below, not as a further layer on the scorecard.

AROPE by age group, Greece, 2015–2025

Every other age group's line bends down after the crisis years; the 65+ line reverses course from 2020 onward

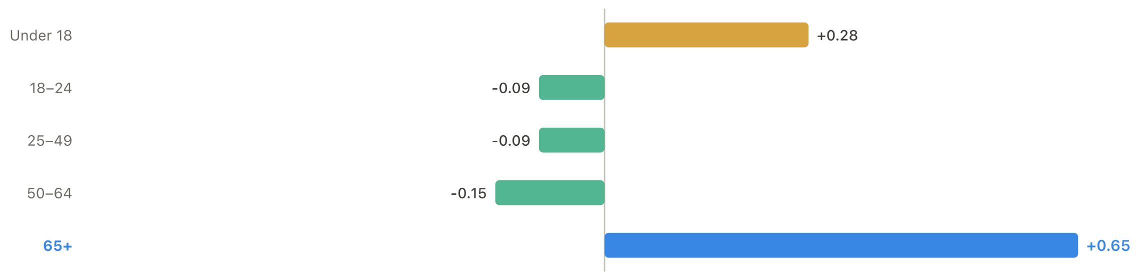


Source: Eurostat, ilc_peps01n, age breakdown — 2015–2025

Core finding **Young adults improved from an exceptionally bad starting point; people 65+ moved the other way.** AROPE for ages 18–24 fell from 43.6% (2015) to 33.2% (2025) — still the highest of any age group, but a real recovery, not a ten-year decline. Every other working-age group (25–49, 50–64) also improved over the same span. AROPE for 65+ did the opposite: 17.5% (2015) → 19.4% (2020) → 27.7% (2025), with the sharpest single-year jump between 2024 and 2025 (24.9% → 27.7%). Children under 18 sit in between: a long decline from 37.7% (2015) followed by a recent reversal, 27.9% (2024) → 29.6% (2025).

Why the national AROPE rate rose in 2025: a shift-share decomposition

Each age group's own contribution to the +0.6-point national change, 2024→2025, split into rate change (this bar) and population-composition shift (not shown — negligible for every group)



Population shares from Eurostat's own AROPE persons-in-thousands columns, not assumed — 2024–2025

This is not demographic aging. It's a rate increase concentrated in one group, large enough to outweigh real improvement everywhere else. Decomposing the national AROPE change into a "population composition shifted toward older, higher-risk groups" effect and a "each group's own rate changed" effect: the composition effect is +0.02 points — essentially zero. The rate effect is +0.60 points — almost the entire story. Broken down by group, every working-age group's own rate *improved* (18–24: −0.09pt contribution; 25–49: −0.09pt; 50–64: −0.15pt) and children's own rate change added +0.28pt. The 65+ group alone contributed **+0.65 points** — more than the entire net national change on its own. Population aging isn't driving this; a real deterioration specific to older people is, and it's large enough to be masking genuine progress in the rest of the population.

People 65+: AROP, deprivation, and AROPE over time

YEAR	AROP, 65+	DEPRIVATION, 65+	AROPE, 65+	AROPE, WHOLE POPULATION
2015	13.7%	7.1%	17.5%	32.4%
2020	13.0%	10.6%	19.4%	27.4%
2023	17.6%	12.3%	23.9%	26.1%
2024	18.8%	12.8%	24.9%	26.9%
2025	**20.9%**	**14.1%**	**27.7%**	**27.5%**

Very-low-work-intensity, AROPE's third component, is structurally undefined for this age group (Eurostat publishes no 65+ breakdown for it at all) — the 65+ increase can only be coming from income poverty, material deprivation, or their unobserved overlap.

Among people 65+, the burden is concentrated specifically among women and people living alone. Greek women 65+ saw AROPE rise from 18.7% (2015) to 30.9% (2025), a 12.2-point increase; men 65+ rose from 15.9% to 23.6%, 7.7 points — the gap between them widened from 2.8 points to 7.3 points over the decade. At EU27 level over the same years, both groups rose only modestly by comparison — women 65+ from 20.6% to 21.2% (+0.6 points) and men from 14.7% to 15.8% (+1.1 points): Greece's elderly women are diverging sharply from the EU pattern, not simply worse off in the same direction as everyone else. By household type, the split is sharper still: Greek single-person 65+ households saw AROPE rise from 23.0% (2015) to **38.7%** (2025), a 15.7-point increase — the largest movement of any group or subgroup checked anywhere in this report. Greek 65+ couples rose from 18.2% to 26.2%, 8.0 points — real, but roughly half the single-person increase. Cross-country context sharpens this further: Greek single-elderly households started *below* the EU27 average in 2015 (23.0% vs. 26.9%) and are now *above* it (38.7% vs. 30.0%) — a genuine reversal, not a persistently bad starting position drifting further behind in parallel with the rest of the EU.

Methods, robustness and caveats

Why AROP goes back further than AROPE and deprivation in this section. Age-specific AROP is shown from 2003 because the series is methodologically continuous — no legacy/revised split exists for it at all. Age-specific AROPE and material deprivation are shown from 2015 only, because Eurostat's revised age-breakdown series begins there; older legacy values are not spliced in, because a direct overlap check (legacy vs. revised, 2015–2020) shows age-specific breaks too large to trust — up to 7.3 points for 18–24 and 5.4 points for 65+, well above the roughly 1–3-point gap this project already tolerates when splicing the whole-population AROPE series elsewhere. The longer AROP series adds useful context without changing this section's main finding: Greek 65+ AROP was 29.4% in 2003, fell steadily to a low of 11.6% in 2018, then rose for seven consecutive years to 20.9% in 2025 — still below its 2003 level, so the current elderly deterioration is a substantial partial reversal of an earlier, larger improvement, not a return to unprecedented territory. That longer arc is background, not this section's headline; the 2015–2025 window above remains the load-bearing comparison, since it's the span where AROP, AROPE, and deprivation can all be read on one consistent basis together.

Methods, robustness and caveats

Method, and three things checked and found not to be usable, reported rather than worked around. Age groups follow Eurostat's own standard EU-SILC breakdown (Y_LT18, Y18–24, Y25–49, Y50–64, Y_GE65), not a custom bucketing. Household-composition figures use Eurostat's own `A1_GE65` (one adult, 65+) and `A2_GE1_GE65` (two adults, at least one 65+) categories directly — a genuine Eurostat dimension, not inferred. Three things were checked and deliberately not used, rather than approximated: (1) *sex within single-elderly households* — Eurostat's household-composition tables have no sex dimension at all, and the single-female/single-male categories in the same tables aren't age-restricted, so combining them with the 65+ household category would fabricate a cross-tab Eurostat doesn't publish; (2) *tenure by age* — the housing-cost-overburden-by-tenure table has no age dimension, and the one table that does cross household type with tenure is a population-distribution table (what share of each household type lives in each tenure category), not an overburden-rate table, so it cannot support a claim like "single elderly owners face X% overburden"; (3) *formal sampling uncertainty* — Eurostat's dissemination API exposes no standard error or reliability flag for any of the cells used here (checked directly in the raw API response, not assumed absent). The qualitative caution that stands in for a formal one: Greece's single-65+ population, while real, is a modest 273,000 people as of 2025, so single-year movements for this and smaller subgroups carry more normal EU-SILC sampling variance than the national headline rate. The decade-long trend (2015→2025) is this section's load-bearing claim; the sharp 2024→2025 jump corroborates it but is not, on its own, independently definitive. Full data and the shift-share derivation: `scripts/39_age_breakdown_arope.py`; output tables `age_breakdown_arope.csv`, `age_breakdown_arop.csv`, `age_breakdown_deprivation.csv`, `age_breakdown_low_work_intensity.csv`, `age_breakdown_eu_arope_rank_2025.csv`, `age_breakdown_shiftshare_decomposition.csv`, `age_breakdown_65plus_by_sex.csv`, `age_breakdown_household_arope.csv`, `age_breakdown_household_arop.csv`, `age_breakdown_legacy_revised_overlap_check.csv`. Full narrative log in `docs/publication_strategy.md`.

DOES GREECE'S SUBJECTIVE POVERTY MOVE WITH THE ECONOMY THE WAY OTHER COUNTRIES' DOES?

A different cut on the same "scarring" idea: instead of levels, look at year-over-year *changes*. For each country separately, its own year-to-year swing in a given predictor is correlated against its own year-to-year swing in subjective poverty — asking whether a country's reported hardship reacts to its own economic news the way the pooled EU pattern (Movement 2's Methods) suggests it should.

YEAR-OVER-YEAR CHANGE IN...	GREECE'S OWN CORRELATION	GREECE'S EU RANK	TYPICAL (MEDIAN) EU COUNTRY
AROP income-poverty rate	+0.71	3rd of 27	+0.12
Real GDP per capita	+0.13	7th of 27	−0.27
Material deprivation	+0.27	19th of 27	+0.40
Unemployment rate	−0.12	21st of 27	+0.01
Housing cost overburden	−0.17	22nd of 27	+0.12
Inability to cover an unexpected expense	+0.02	24th of 27	+0.58
Arrears	−0.34	27th of 27 (lowest)	+0.26

Descriptive comparison, not a causal test **Greece's subjective poverty tracks its own AROP income-poverty rate unusually closely — and almost nothing else.** On year-to-year changes in AROP, Greece is 3rd-most-tightly-tracking of 27 countries. On every material-hardship measure tested — unemployment, housing costs, deprivation, arrears, unexpected-expense capacity — Greece sits in the bottom third of the EU, and on arrears its correlation is not just weak but the single most negative of any country (more arrears in a given year is, if anything, associated with a very slightly *lower* subjective-poverty reading that year). Most EU countries' subjective poverty rises and falls with these year-to-year swings much more than Greece's does. Read alongside the previous subsection, this fits a consistent picture: Greek subjective poverty behaves less like a reaction to each year's specific economic news and more like a persistently elevated *level* — consistent with the scarring finding above (a hard economic condition), and distinct from swing-reactive hardship. The financial-pessimism finding below is a separate, more heavily caveated result and shouldn't be read as the same kind of evidence as this one.

Methods, robustness and caveats

Exploratory lead, not a core finding **Does that sensitivity pattern itself relate to the size of a country's unexplained gap?** Tested directly: for each of the nine predictors above, does a country's own Δ-sensitivity correlate with the size of its Model C leave-one-out residual (the legacy section) across all 27 countries? With nine tests run together, a false-discovery-rate correction (Methods) is the right way to read the p-values, not the raw ones.

Under that correction, eight of the nine show no reliable relationship, and the ninth — real GDP — survives, but only just: countries whose subjective poverty is less connected to (or moves opposite to) real GDP swings tend to have larger unexplained gaps (r=+0.52, raw p=0.006, FDR-adjusted p=0.050), and this holds almost unchanged excluding Greece (r=+0.50, raw p=0.009), so it isn't just Greece driving its own correlation. Greece's own real-GDP sensitivity (mildly positive rather than the expected negative, ranked 7th of 27) fits that broader pattern rather than being a one-off. Arrears looked similarly related at first (r=−0.46, raw p=0.016) but fails the FDR correction outright (adjusted p=0.070) and weakens further once Greece is excluded (p rises to 0.058) — that one is reported as "Greece happens to be unusual on two separate things at once," not a finding. With only 9–10 year-over-year observations feeding each country's own correlation, this whole subsection is more descriptive than confirmatory — a different, weaker standard of evidence than the panel regressions used elsewhere in this report, and labeled as such.

HOW PESSIMISTIC HOUSEHOLDS ARE ABOUT THEIR OWN FUTURE

So far: Greece is uniquely still below its own income peak (hard data), and its subjective poverty barely reacts to yearly hardship swings (a descriptive pattern). The scorecard's Model E adds a third, different kind of variable — not a material condition at all, but how Greek households themselves expect their finances to change.

Descriptive check — read with the caveats below **Greece is also the single most pessimistic country in the EU about where its own households' finances are headed, and this remains true regardless of which unemployment measure the model uses.** Using Eurostat's household financial-expectations survey (how households expect their own finances to change over the next 12 months), Greece scores worst of 35 countries measured, 11 points clear of the next-most-pessimistic (Slovenia). Added to Model C (headline unemployment), it improves the fit (R^2 0.892→0.904), is itself statistically significant (p=0.021), and knocks Greece from the largest cross-country gap in the EU down to 10th of 27. But financial pessimism is not fully independent of the labor-market finding above: added to Model C-LTU instead of Model C, its own coefficient shrinks and its significance weakens to p=0.062 — no longer significant at the conventional 5% threshold. Financial pessimism remains real and important as an outcome-adjacent signal, but some of its apparent explanatory power overlaps with long-term unemployment rather than standing fully apart from it.

Methods, robustness and caveats

Caveat that matters more here than anywhere else in this report: "how do you expect your own household's finances to change" is a subjective survey question, much closer in kind to "how difficult is it to make ends meet" (this report's outcome variable) than any material variable tested elsewhere. This result is better read as "current difficulty and pessimism about the future are closely related sentiments that move together in Greece more than in the rest of the EU," not as an independent economic condition that explains the gap. It reframes part of the residual rather than resolving it — from "an unexplained material gap" to "Greece is unusually pessimistic about its own future, in a way partly separable from current material conditions." **Household saving rate** was also tested as a possible wealth-depletion signal (Greece is 2nd-lowest of 34 countries, persistently negative almost every year 2015–2024) but added no independent explanatory power once financial expectations was already in the model — consistent with the two being correlated rather than saving rate carrying separate information, and with this report's recurring limitation that national averages can't see who, within a country, is actually depleting savings. Independent corroboration for the pessimism finding comes from outside Eurostat entirely: a University of Macedonia survey for the Greek think tank diaNEOsis (April 2016, n=1,348 per the results file, reported as "1,300" in diaNEOsis's own summary text, ±2.7%) found 74% of respondents expected their own financial situation to worsen over the next year (7% expected improvement) and 81.5% reported difficulty meeting basic household needs — a different survey house, a different year, and different question wording, landing on the same extreme pessimism.

Methods, robustness and caveats

This pessimism finding should not be read as evidence of a national temperament — and the trust story behind it is older than the crisis, not created by it. A separate literature on institutional trust specifically, not modeled here (see Methods for why), points to a more structural reading, in four parts. **Pre-existing condition:** Greece did not enter the 2010s as a high-trust society that then collapsed. The OECD's own 2026 country profile, using Eurobarometer data back to 1995, describes pre-crisis trust in national government as "not particularly high" — averaging 44% across the pre-2008 years, with a single anomalous peak above 50% in 2004 (the Athens Olympics). **Crisis shock:** that already-middling baseline was then hit hard — trust fell to a historical low of 7% in 2012 — the same year this report's own migration data (above) marks as the peak of crisis-era emigration from Greece, independent corroboration that 2012 was among the most acute years of the crisis, distinct from the later, deeper GDP trough of 2020 (above). **Aftermath:** recovery has been partial and fragile — trust peaked again at 37% in both 2015 and 2020, never regaining even the middling pre-2008 average, and has fallen again since 2023 (32% down to 24%) despite genuinely strong recent macroeconomic performance, a disconnect the OECD's own account treats as puzzling rather than explained. Ervasti, Kouvo & Venetoklis (2019), using European Social Survey data for 2002–2011, add a further distinction: it was specifically trust in political and impartial institutions that the crisis damaged, while interpersonal trust, trust in other people, held steady — the crisis didn't make Greeks generally more distrustful people. **Interpretation:** read together, institutional trust in Greece is better understood as long-standing background weakness that the crisis deepened and made politically salient, not a clean before/after story and not something this report treats as an independent cause of the pessimism finding above — context and mechanism, not a variable that explains it away. Economou et al. (2014) separately link low institutional and interpersonal trust to worse mental health outcomes during the crisis in a nationwide Greek study. None of this is tested as a variable in this report's own models. **Evidence role:** social/demographic scarring context, the same role migration plays above — it helps explain why a macro recovery hasn't yet felt like one, not an independent, modeled cause of the pessimism finding.

WORKING THE MOST HOURS, EARNING THE LEAST PER HOUR

Everything so far treats employment status as a dividing line — unemployed and long-term unemployed people carry most of this report's material-scarring story. This section asks a sharper question: does having a job actually solve the problem? For salaried Greek workers specifically — removing joblessness from the picture entirely — the answer is no.

Salaried workers only, 2025: official poverty measures vs. reported hardship

	GREECE	EU27	GREECE'S EU RANK
AROP (income poverty)	5.7%	6.5%	17th of 27 — unremarkable
AROPE (poverty or exclusion)	12.5%	9.2%	3rd of 27
Subjective poverty	**59.5%**	**13.6%**	**1st of 27**

Core finding **Greek salaried workers rank a completely ordinary 17th of 27 EU countries on official income poverty — and 1st, by the widest margin in Europe, on reported difficulty making ends meet.** This shows the puzzle persists among people who are working: their income poverty rate looks unremarkable by the EU's own official yardstick, and nearly six in ten of them still say they struggle to make ends meet, more than four times the EU average. Whatever is driving this report's central gap, it cannot be explained away as "it's really an unemployment story" — it survives, undiminished, among people who are working.

One candidate explanation: maybe Greek salaried workers are working unusually long hours for unusually little pay per hour — not a headline salary problem, but an effort-versus-reward one.

Actual weekly hours vs. compensation per hour worked, 2024

COUNTRY	WEEKLY HOURS, ALL EMPLOYED	HOURLY COMPENSATION (PPS)	WORK-EFFORT SQUEEZE (EU=100)
Greece	**39.8**	**14.2**	**230.5 — highest in the EU**
Bulgaria (2nd)	39.0	17.4	184.3
Poland (3rd)	38.9	17.6	181.7
EU27 average	36.0	29.6	100.0
Luxembourg (lowest)	35.7	46.2	63.5 — lowest in the EU

Source: Eurostat, lfsa_ewhan2 (hours) & nama_10_lp_ulc (compensation per hour, PPS) 2024

Core finding **Greece works the longest hours in the EU and gets paid the least per hour for it — on every way of measuring hours tested.** Actual weekly hours across all employed people: 39.8, highest of 27 (EU average 36.0). Full-time weekly hours: 41.1, also highest (EU 38.8). National-accounts annual hours per employee: 1,900, 4th-highest (EU 1,547). Hourly compensation, adjusted for purchasing power: €14.2-equivalent, the lowest of any EU country (EU average 29.6). Combined into a single "work-effort squeeze" index — relative hours worked divided by relative hourly pay, EU=100 — Greece scores 230.5, the highest in the EU by a wide margin. The pattern holds even restricting to salaried employees only, ruling out Greece's comparatively large self-employed population as the explanation: salaried weekly hours alone still rank 7th-highest in the EU.

The hourly-pay side of this isn't just a cross-country gap — it's also a story of lost ground. Real hourly compensation (inflation-adjusted, indexed to each country's own 2008 level) remains **27.8% below** its 2008 level in Greece as of the latest data. Combined with the hours finding, the picture is not "Greeks don't work enough" or "Greeks are paid what their output is worth" — it's a country where an hour of work now buys noticeably less than it did before the crisis, while people are working more hours than anywhere else in the EU to compensate.

Methods, robustness and caveats

What this is, and what it isn't. Added to this report's richest cross-country model (Model C-LTU), the work-effort squeeze is statistically significant even after correcting for testing multiple candidates together (FDR-adjusted p=0.003), and moves Greece's out-of-sample gap from +3.9 points to −3.5. But tested directly for whether it behaves like long-term or cumulative unemployment — tracking a country's own year-to-year movement, not just its position relative to other countries — it does not: adding country fixed effects makes its coefficient non-significant (p=0.34), first-differencing on year-to-year change makes it non-significant too (p=0.73), and within Greece's own 15-year time series the squeeze measure and subjective poverty are uncorrelated (r=−0.03, p=0.92). **Evidence role:** this is a genuine, stable, cross-country structural fact about who Greece is — not a dynamic mechanism explaining what changed for Greece over time. Note that the same limitation applies to cumulative excess unemployment: it too is supported predominantly between countries, so the contrast here is between two structural facts, not between a static one and a dynamic one. It is not added to the scorecard alongside long-term and cumulative unemployment for that reason, and is kept here as strong supporting evidence — on its own, next to AROP, it closes 13.8 points of Greece's 52.6-point average (2015–2024) raw AROP gap — for the same "material conditions, not mood" argument the rest of this section makes. Full derivation, the salaried/self-employed decomposition, the FDR-corrected candidate battery, and the within-country robustness checks are reproducible from `scripts/43_work_effort_squeeze.py`, logged in `docs/publication_strategy.md`.

WHY SOME COUNTRIES FEEL ALMOST EXACTLY AS POOR AS THEY MEASURE

The three findings above all look at what's unusual about Greece in isolation. This one flips the lens: instead of only asking what makes Greece unusual, look at the EU countries where subjective poverty and the broader poverty-or-exclusion rate (AROPE, Movement 2) line up almost exactly — Italy, France, Portugal, Hungary, Croatia, Cyprus, Belgium, and Slovenia, each within 2 points, versus Greece's 39.7-point gap. If these countries share some trait Greece lacks, that's a useful contrast.

Greece vs. the eight closest-to-aligned EU countries, most recent year available

	UNEMPLOYMENT	MATERIAL DEPRIVATION	HOUSING COST OVERBURDEN	ARREARS	BELOW OWN INCOME PEAK	FINANCIAL PESSIMISM
Greece	10.1%	14.0%	28.9%	42.8%	11.2%	−43.2 (worst of 27)
8-country average	5.5%	4.7%	5.6%	8.2%	0.0%	−7.4

Descriptive comparison, not a causal test These eight countries are not a uniform group — individually, several score worse than Greece on single measures (Hungary's material deprivation, Cyprus's arrears). What none of them do is combine several kinds of strain at once the way Greece does: Greece ranks 1st or 2nd of 27 on housing cost overburden, arrears, and distance below its own income peak, and 3rd on material deprivation and inability to cover an unexpected expense — and every one of the eight comparison countries has fully recovered to its own historical income peak, while Greece remains 11.2% below its. None of them approaches Greece's degree of pessimism about the future, either. The pattern this suggests: it isn't any single hardship measure that pushes a country's subjective poverty far above what its AROPE rate implies — it's several kinds of strain, plus unhealed income scarring, plus collapsed expectations, stacking at once. That's a description of what's different about Greece, not a causal test of it, and it draws on the same small set of variables already discussed above rather than an independent check.

MOVEMENT 4

The objective-only model

The question: How much of Greece's reported hardship is associated with observable conditions that are *not* near-restatements of the outcome itself?

A model that excludes anything resembling the answer

Earlier versions of this analysis leaned on two predictors — being behind on bills, and having no buffer for an unexpected expense — that are close to the outcome itself. Someone who cannot meet an unexpected expense is, in most ordinary senses, already having difficulty making ends meet. Models containing them fit better and explain less. This movement reports the result from a specification that excludes them by construction.

What is in and what is out. The specification is fixed in advance: severe material and social deprivation, housing-cost overburden, long-term unemployment, income (actual individual consumption per capita in PPS), years of real wages below their own 2008 level, and accumulated excess unemployment, with year fixed effects, unweighted, on 269 country-years across all 27 member states. **Excluded by construction** as proximate to the outcome: arrears, inability to meet an unexpected expense, and financial expectations. Greece is left out of fitting and then predicted, so the number below is what a model that has never seen Greece gets wrong about Greece.

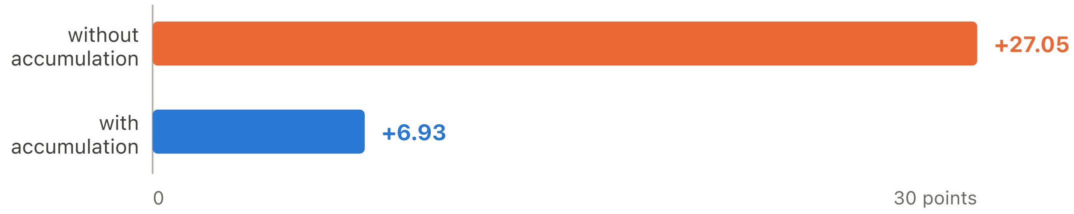
Without cumulative unemployment exposure, the objective-only model fails badly on Greece. With deprivation, housing costs, long-term unemployment, income and the real-wage record — note that this already includes one history variable, the years real wages have been below their own 2008 level, so this is not a purely current-conditions model — Greece's out-of-sample residual is **+27.05 points**, the **largest of all 27 member states**. On observable current circumstances alone, Greek households report far more hardship than any European comparison predicts.

Adding the accumulated record of unemployment narrows the gap to a seventh of that. Adding accumulated unemployment exposure to the objective-only specification reduces Greece's leave-out residual **from approximately 27 to 7 points** (+6.93), moving Greece from 1st to **3rd of 27**. Greece nevertheless remains the third most under-predicted country in the Union: accumulated material history substantially narrows the puzzle but does not resolve it.

That +27.05 is not an artefact of this particular predictor set. The earlier, simpler specification — unemployment, income, deprivation and AROP — left Greece **25.6 points** under-predicted and likewise **1st of 27**. Two differently built models agree on the finding that matters here: on observable conditions without a measure of accumulated labour-market history, Greece is the most under-predicted country in the Union.

The objective-only model, with and without accumulated exposure

Greece's out-of-sample residual, percentage points. Both bars from the identical 269-country-year sample and the identical leave-one-country-out procedure.



rank 1st of 27 → 3rd of 27

a comparison of two model specifications, not a decomposition

How to read that reduction. It is a **model comparison, not a causal decomposition**. Twenty points is the difference between two specifications estimated on the identical 269-row sample with the identical leave-one-country-out procedure; it is not a quantity of hardship attributable to accumulation. The coefficient on accumulated exposure is +0.281, stable in sign across all 27 leave-one-country-out folds, with a variance inflation factor of 3.3 alongside long-term unemployment in the same model. Evidentiary status: **post-selection robustness, not independent confirmation** — accumulated excess unemployment was discovered on this same panel, so this validates a pre-specified model against the data that produced the hypothesis.

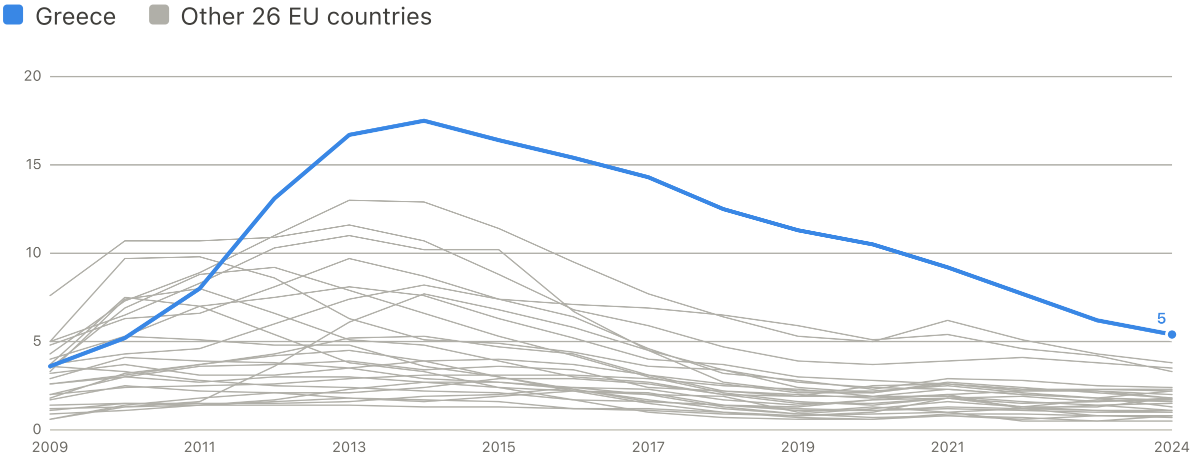
On the smaller residuals in earlier specifications. Specifications that admitted arrears and unexpected-expense capacity produced visibly smaller Greek residuals than the +6.93 reported here — in the richest of them the residual approached zero. Those results are not being hidden: they are reported in full in the statistical appendix and are **superseded**, not refuted. They are proximity-sensitive, in the precise sense that their apparent explanatory success comes from predictors that restate the outcome. A model that explains difficulty making ends meet using arrears has not explained much. The headline figure is therefore the larger, more honest +6.93 from the objective-only specification, and the earlier ladder is kept in the appendix as a legacy record.

SUPPORTING EVIDENCE: UNEMPLOYMENT LASTED YEARS, NOT MONTHS

Pay didn't recover. Neither, for a long time, did jobs. Headline unemployment tells you how many people are out of work right now. It doesn't tell you whether the labor market has actually started absorbing them back in, or whether the same people have simply stayed unemployed year after year. Long-term unemployment — the share of the labor force out of work for 12 months or more — tells you that. It turns out to be the strongest single variable tested anywhere in this report.

Long-term unemployment rate (12 months or more, % of active population)

Every EU country's line is shown; Greece is the only one still well above the rest of the EU in 2024



Source: Eurostat, une_ltu_a

2009–2024

Supporting evidence **Greece's long-term unemployment rate has fallen a long way from its crisis peak — but it is still, by a wide margin, the EU's highest.** It peaked at 17.5% of the labor force in 2014 (meaning roughly three-quarters of all unemployed Greeks, at that point, had been out of work for a year or more) and has fallen steadily since, to 5.4% in 2024. That's real progress. But 5.4% is still the highest in the EU — next is Spain at 3.8%, then Slovakia (3.5%) and Italy (3.3%); most of the EU is under 2%. Headline unemployment has recovered a lot faster than this has.

It tracks Greek subjective poverty more tightly than any other variable tested — at every time scale. Level correlation: $r=0.93$ ($p<0.0001$). Year-to-year changes: $r=0.92$ ($p<0.0001$) — unlike almost every other variable in this report, which mostly show weak or non-significant year-to-year correlations even when their level correlation is strong. Detrended: $r=0.86$ ($p<0.0001$). All three survive false-discovery-rate correction comfortably (Methods), and long-term unemployment is now the single highest-ranked variable in the correlation table in Movement 2.

Swapped into the cross-country model in place of headline unemployment, it substantially changes Greece's story. Model C-LTU (the legacy section) lifts the model's fit from R^2 0.89 to 0.91, and cuts Greece's out-of-sample gap from 11.6 points to 3.9 — a fall of roughly two-thirds. Greece is no longer the EU's largest unexplained outlier under this specification:

The six largest out-of-sample gaps, before and after swapping in long-term unemployment

MODEL C (HEADLINE UNEMPLOYMENT)	GAP	MODEL C-LTU (LONG-TERM UNEMPLOYMENT)	GAP
1. Greece	11.6	1. Cyprus	8.9
2. Luxembourg	10.7	2. Luxembourg	8.3
3. Cyprus	8.1	3. Portugal	6.6
4. Portugal	8.0	4. Czechia	5.0
5. Italy	7.4	5. Belgium	4.3
6. Slovakia	6.1	6. Greece	3.9

Full 27-country ranking for both specifications: `scorecard_loo_C_baseline.csv` and `scorecard_loo_C_LTU_swap.csv`.

Why this is a replacement, not an addition — and why that choice matters. Long-term unemployment and headline unemployment are, unsurprisingly, highly correlated (r=0.91 across the full 27-country panel; r=0.998 for Greece's own time series alone, since by the mid-2010s most of Greece's unemployment *was* long-term). Adding long-term unemployment *alongside* headline unemployment in the same regression (tested as robustness) improves the fit slightly further (R^2 0.92, Greece's gap down to 2.3) but destabilizes headline unemployment's own coefficient — it flips to a counterintuitive negative sign and loses statistical significance, a textbook symptom of collinearity (variance inflation factor ≈8–9 for both variables together). Replacing headline unemployment with long-term unemployment instead avoids that problem entirely, and long-term unemployment's own coefficient is remarkably stable: refitting the model 27 times, once with each country left out in turn, its coefficient never changes sign and is never less significant than p<0.001 — including the refit that excludes Greece itself. That stability is why this report treats long-term unemployment, used as a replacement, as the preferred specification, and the additive version as a robustness check only.

Does this compete with the other variables? Partly. The GDP scarring stock adds nothing beside LTU (p=0.61), while financial expectations weakens from p=0.021 to p=0.062. The evidence supports overlapping markers of prolonged hardship, not three independent mechanisms and not a proven causal chain.

SUPPORTING EVIDENCE: HOW THE DURATION MEASURES WERE SELECTED AND CHECKED

Long-term unemployment measures how many people are currently out of work for a year or more. It doesn't measure how much accumulated exposure a whole economy has carried since the crisis began. A country that had ten brutal years and has now largely recovered, and a country that has had ten mildly-bad years without ever recovering, can show the same headline or long-term unemployment rate today while carrying very different amounts of accumulated hardship. This section tests that accumulated-exposure idea directly: for every country and year since 2009, unemployment above that country's own baseline rate is summed, year over year (floored at zero, so later improvement never cancels out earlier damage). The result is each country's running total of "excess unemployment-years" carried since the crisis began — and, as the robustness checks below show, the effect is really about *sustained* exposure over roughly the past decade, not literally permanent, undiminished accumulation since one specific baseline year.

The gap-closing ladder: from the raw AROP gap to the final model

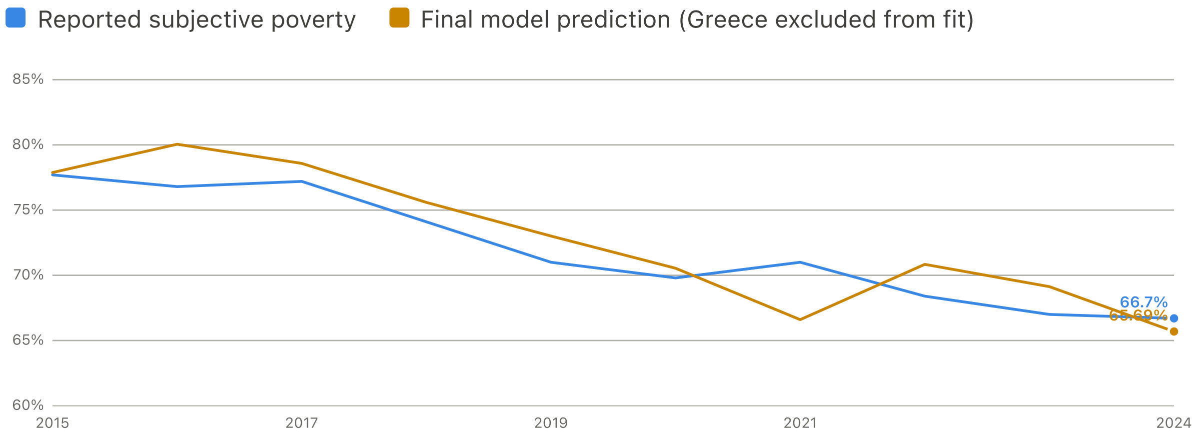
All rows share one common window (2015–2024 averages). Steps 1–2 are raw single-country gaps; steps 3–6 are out-of-sample (leave-Greece-out) model residuals — two different kinds of quantity, deliberately presented as a sequence rather than a single additive decomposition.

STEP	LAYER	GREECE'S GAP	POINTS CLOSED VS. RAW AROP GAP
1	AROP raw gap (2015-2024 avg) (raw, single-country)	+52.6	—
2	AROPE bridge (2015-2024 avg) (raw, single-country)	+41.5	+11.1
3	Basic model: AROP + income + deprivation + headline unemployment (Model A) (regression, OOS)	+25.6	+27.0
4	+ housing, arrears, unexpected-expense capacity (Model C) (regression, OOS)	+11.6	+41.0
5	Headline unemployment -> LTU (Model C-LTU) (regression, OOS)	+3.9	+48.7
6	**+ cumulative excess unemployment (final model) (regression, OOS)**	**-0.8**	**+53.4**

Supporting evidence **In the earlier proximity-sensitive screening, cumulative excess unemployment since 2009 was the strongest single addition tested.** Added on top of Model C-LTU, it lifts the fit from R^2 0.914 to 0.930 and moves Greece's out-of-sample gap from 3.9 points to −0.8 *(legacy, superseded)* — the model now slightly *overpredicts* Greece's reported hardship rather than underpredicting it. Once it's the variable being tested, its own coefficient stays positive and stable when the model is refit 27 times, once with each country excluded in turn (range 0.134–0.197, no sign flips anywhere), and remains statistically significant even in the refit that excludes Greece entirely (coefficient 0.164, $p=0.0064$) — that specific result isn't an artifact of Greece's own data point. Whether Greece's data should have counted toward *choosing* this variable over the alternatives in the first place is a separate question, addressed directly below.

Greece: reported subjective poverty vs. the final model's prediction, 2015–2024

Predicted values come from a model fit on the other 26 countries only (Greece excluded), then applied to Greece — a genuine out-of-sample test, not a fit to Greece's own data



Model: C-LTU + cumulative excess unemployment, fit excluding Greece — 2015–2024

The near-zero average residual isn't a single-year fluke. Year by year, the final model's prediction sits close to Greece's actual reported hardship for nine of the ten years shown (residuals from −3.2 to +1.0), with one exception: 2021, where the model underpredicts by 4.4 points, plausibly a pandemic-specific disruption the model isn't built to capture. Removing that one year would make the average fit even tighter, not looser.

Not a substitute for long-term unemployment — a genuine additional layer. Replacing long-term unemployment with cumulative excess unemployment, rather than adding it on top, performs worse (R^2 0.911, Greece's out-of-sample gap 4.3) — so this isn't long-term unemployment relabeled. Used together, both coefficients stay significant and variance inflation factors stay reasonable (max 4.5, well below the usual concern threshold). Cumulative excess unemployment captures something long-term unemployment's snapshot alone does not: how much accumulated exposure the whole economy carried, not just how many people are currently stuck.

Supporting, not central: years spent below the 2008 real-wage level. A related duration measure — how many consecutive years a country's real wages stayed below its own 2008 level — is independently significant on its own ($p=0.0045$) and passes the same leave-one-country-out stability check (coefficient range 0.31–0.54, always significant, including when Greece itself is excluded: coefficient 0.31, $p=0.0036$). The equivalent GDP-based duration and direction measures tested (years below 2008, years below own historical peak, longest continuous shortfall streak, cumulative negative-growth years) all came back non-significant ($p=0.13$–0.89) — duration matters here specifically for wages, the number that shows up in a payslip, not for GDP, the more abstract aggregate. Combined with cumulative excess unemployment, both variables stay significant, but within Greece's own time series the two move almost in lockstep (correlation 0.95, versus 0.54 across the full panel) — too entangled, in Greece's case specifically, to credibly separate as two independent mechanisms. Kept here as robust supporting evidence for the same underlying story, not stacked on the scorecard as a further, independent layer.

What this section ruled out, rather than confirming. The most narratively obvious candidate — a cumulative shortfall in the AROP poverty threshold itself, tracking how far behind Greece's income-poverty line fell relative to its own pre-crisis path — was tested first and came back statistically null (p=0.291), despite being exactly the kind of "the ruler itself has been shrinking for a long time" measure Movement 2 leads with. It's reported here for completeness, not folded quietly out of the record: a cumulative-threshold effect was a reasonable hypothesis and the data doesn't support it.

Method: unemployment and long-term unemployment excess are each computed against the country's own rate in a 2009 baseline year (the earliest year with near-complete 27-country coverage for both series), summed year over year and floored at zero so a country cannot "bank" credit from a good year against a bad one earlier. Full derivation, every candidate tested (including the eight-variable original battery and the duration/direction battery), the replacement test, both leave-one-out stability checks, and the year-by-year residual table are reproducible from `scripts/38_cumulative_hardship.py` and logged in full in `docs/publication_strategy.md`.

MOVEMENT 5

What the result is not

The question: A narrowed gap invites a reading the evidence does not support. This movement states what the finding rules out, what remains unresolved, and which alternative explanations were tested and failed.

5

The association is between countries, not within them

The evidence is predominantly between countries. Splitting the accumulated-exposure coefficient into its between-country and within-country parts gives **+0.332 between** (p<0.0001) and **−0.076 within** (p=0.692). Countries that accumulated more labour-market exposure report more hardship. The panel does **not** establish that as a given country accumulated more exposure, its own reported hardship rose.

Three further tests point the same way, and they are not independent — each uses the same scarce within-country variation. Country and year fixed effects together give −0.039; first differences, which ask whether a year's newly accumulated exposure tracks that year's change in reported hardship, give −0.003 (p=0.98); country-specific linear trends give +0.049.

What the within-country null does and does not establish. The within estimate's standard error is 4.8 times the between estimate's, with a 95% interval of [−0.449,

+0.298]. A within–between equality test returns p=0.058, which *fails to reject* equality and is **not** evidence that the two are equal — the point estimates remain materially different. The within component is therefore **inconclusive under available power**, not unsupported with adequate sensitivity. This analysis can neither establish a within-country dynamic effect nor rule one out. Accumulated exposure should be described as a **between-country scarring marker**, never as a mechanism, and the sentence "as exposure accumulated, hardship increased" is not supported by anything here.

Stability, by contrast, is not in doubt. Dropping any single country moves the coefficient by at most 12.7% (Cyprus); dropping *Greece* moves it by 0.8%, so Greece is not driving its own result. Accumulated exposure remained supported across all bootstrap variants (p_MC=0.0005 in the primary specification; worst robustness result p_MC=0.0070 across alternative weight schemes and seeds).

TWO ALTERNATIVE EXPLANATIONS WERE BUILT, PRE-REGISTERED, AND FAILED

The comparative (synthetic-control) design failed its pre-registered gates. A synthetic Greece was assembled from countries resembling pre-crisis Greece, with the donor pool, windows and pass criteria fixed in writing beforehand. It failed four of six gates. The decisive failure was not statistical but substantive: the synthetic unit that best matched Greece's pre-crisis *outcome* had roughly **half Greece's income and three times its deprivation**. A comparison unit that is not comparable cannot support an estimate, so **no post-crisis synthetic-control effect is interpreted anywhere in this report**.

Breadth of deterioration is descriptive only. The multi-domain breadth measure from Movement 3 — on which Greece ranks first in the Union — was also tested as a predictor. Added to the objective-only model it does **not** improve prediction: Greece's residual moves the wrong way, from +6.93 to +10.39, making Greece the single most under-predicted country. Its conditional coefficient reverses sign, a stable reversal consistent with suppression or construct overlap, and it is left uninterpreted. The breadth finding stands as a description of how widely Greece deteriorated; it is **not** an additional explanation.

Both failures are reported because a screening record that keeps only its successes is not a record. The same applies to the nulls already noted: income inequality, household debt, welfare-transfer effectiveness, and the cumulative shortfall in the poverty threshold itself all failed to add explanatory power once the model included what it includes.

§

Superseded: the earlier model ladder

Where the earlier results went, and why. Previous versions of this report built their central argument on a model ladder that admitted arrears and unexpected-expense capacity — predictors close enough to "difficulty making ends meet" that their explanatory success is partly circular. Those specifications produced **visibly smaller Greek residuals** than the +6.93 reported in Movement 4; the richest of them brought the residual to roughly zero, and a nested select-then-fit procedure gave +2.70 at rank 19 of 27. **They are superseded,**

not refuted, and the smaller numbers are precisely the reason. A model that explains difficulty making ends meet using arrears has not explained much. The full ladder, its tables, its scorecard and its diagnostics are retained in the statistical appendix so that nothing is hidden and the reasoning that retired them stays auditable. **No residual from that ladder is a current headline estimate.**

MOVEMENT 6

Discussion

The question: What does this evidence support, what does it leave open, and what should not be concluded from it?

The Greek result is materially grounded and it is not fully explained. Those two statements are both true and neither replaces the other. A model containing nothing that restates the outcome leaves Greece 27 points under-predicted without cumulative unemployment exposure; adding the accumulated record of unemployment brings that to roughly 7. Greece nonetheless remains the third most under-predicted country in the Union, and the association is carried by differences *between* countries rather than by change within Greece over time.

What that supports is a specific and limited claim: Greek households' reports of hardship track a measurable material history, and a country that spent a decade accumulating labour-market damage looks much less anomalous than one judged on this year's conditions. What it does not support is any statement that the puzzle is solved, that the mechanism is identified, or that accumulation caused the reports. The evidence is associational, drawn from country-level aggregates, and cannot speak to what happened inside any household.

On the cultural explanation, the position is narrower than either side of the usual argument. Greece did report unusually high hardship before the crisis, so a stable national component cannot be dismissed. But a stable component cannot produce a widening, and the widening is the largest of any member state. What remains genuinely open is a **financial-domain-specific reporting difference** — a change in how Greek households answer money questions specifically, brought about by the crisis itself. The domain-specificity evidence in this movement rules out generic national gloom. It does not rule that out, and no aggregate design used here can.

Could this still be reporting culture?

One more objection is worth taking seriously, because it isn't dismissible from the numbers already shown: **subjective poverty is a self-report.** If Greek respondents simply describe hardship differently — a harsher cultural baseline, a more pessimistic response style — some of this report's entire starting point could be a survey artifact rather than a material one. This section takes that possibility on directly, with six separate tests, rather than assuming Sections 1–11 already settled it. It is a robustness/context section, not a new headline finding — it exists to check whether the answers already given survive the most obvious skeptical explanation.

WAS GREECE ALREADY THE HIGHEST BEFORE THE CRISIS EVEN BEGAN?

Yes — and this report says so plainly rather than arguing around it. Averaged over 2003–2008, Greece already had one of the EU's highest subjective-poverty levels. That matters: whatever explains the crisis-era rise (Movement 3), it doesn't by itself explain why Greece already stood out beforehand.

But the exact ranking needs a careful caveat, checked directly rather than assumed. EU-SILC, the underlying survey, launched in 2003 with only six countries fully on board — Belgium, Denmark, Greece, Ireland, Luxembourg, and Austria, confirmed directly from Eurostat's own EU-SILC documentation, which also flags disruption in the data as more countries joined through 2005. Comparing Greece against a full 26-country EU panel for those exact years overstates how complete the picture actually was. Checked three different ways — the six countries with genuinely full 2003–2008 coverage, a 25-country panel starting in 2005, and a 26-country panel starting in 2007 — the answer doesn't change. If anything it strengthens: among the six countries EU-SILC actually covered from day one, Greece ranks **1st**, not 2nd.

> Conceded directly **A pre-existing, stable component of Greece's gap cannot be ruled out.** Greece was already near the top of the EU distribution before any crisis-era shock. The remaining tests below ask whether that's the whole story, or just the starting point.

DID GREECE'S GAP GET BIGGER AFTER THE CRISIS, OR HAS IT ALWAYS BEEN THIS SIZE?

A stable cultural trait should produce a stable gap — wide before the crisis, and about the same width after. That is not what happened.

Greece's subjective-poverty-minus-AROP gap, before and after the crisis

	2003–2008	2019–2024	CHANGE
Greece	30.4pt	50.2pt	+19.8pt — largest widening of any EU country
Typical (median) other EU country	—	—	−8.8pt — most countries' gaps narrowed

Core finding **Formally tested with a difference-in-differences model — comparing Greece's own change after 2009 against every other EU country's change over the same years, after removing each country's own baseline and each year's EU-wide shock — Greece's shift is large (+20 points) and does not look like statistical noise.** Checked six further ways, since a test built on a single "treated" country deserves more scrutiny than most results in this report: the size of the shift barely moves whether the crisis is dated to 2009, 2010, or 2011; it holds almost unchanged if any single comparison country is dropped one at a time; it holds, if anything larger, restricting the comparison to Greece's closest peers (Italy, Spain, Portugal, Cyprus, Malta); there's no sign of the gap already widening before 2009 — though the balanced panel only allows testing one pre-crisis year (2007) directly, so this is a necessary check passed, not an exhaustive pre-trend history; and pretending, in turn, that each of the other 26 EU countries was "Greece" and re-running the same test, Greece's own shift is still the largest of all 27, an empirical p-value of 0.037 — an honest way of checking a one-country result that doesn't lean on textbook statistical assumptions built for many treated units. This is a placebo/permutation reference distribution: it reads as a probability only if the other 26 countries are treated as exchangeable stand-ins for Greece. Greece's crisis was not randomly assigned, so this is not randomization inference in the design-based sense. The rank itself is exact; the inferential reading is what carries the assumption. A further check was attempted and is reported here as a **failure**, because reporting it as support would overstate it. The idea was to build a "synthetic Greece" from a weighted blend of countries resembling pre-crisis Greece and watch the two diverge. Over the only sensible pre-crisis window, 2003–2008, it does not work: only six countries have complete data that far back, and the blend never matches Greece in the first place — it sits around 3–8 points where Greece sits at 27–34. A blend that never tracked Greece before the crisis cannot tell you anything by diverging from it after. Matching on just 2007–2008 does give a near-perfect fit, but fitting two data points with twenty-five adjustable weights can readily produce a near-exact fit, so agreement there is weak evidence rather than strong. The conclusions in this section rest on the difference-in-differences result and the placebo test above, not on this.

IS GREECE EQUALLY NEGATIVE ABOUT EVERYTHING, OR SPECIFICALLY ABOUT MONEY?

A generic cultural pessimism should show up everywhere a country is asked to self-report — not just on financial questions. It doesn't, in Greece's case. Note carefully what this does and does not rule out: it rules out *generic* gloom, not a *financial-domain-specific reporting difference*. A sense of unfair treatment by the tax and public-spending system, for instance, would also show up on financial questions and not on life satisfaction, and this test cannot separate it from material hardship.

SELF-REPORTED MEASURE	GREECE'S EU RANK
Can't make ends meet (this report's outcome)	1st of 27, every year checked
Pessimistic about own finances next year	1st of 27, every year checked
General life satisfaction	2nd–6th of 27, never the most extreme

Core finding **Greece is consistently the most negative country in the EU on both financial self-report measures, but not on general life satisfaction — and the gap isn't just about rank, it's about size.** Converting all three to a common, standardized scale (how many standard deviations Greece sits from the EU average, sign-adjusted so "worse" always points the same way), Greece's deviation on the two financial measures is roughly three times larger than its deviation on general life satisfaction. If this were a generic response-style effect — Greeks simply answering surveys more darkly across the board — all three should look similarly extreme. They don't. The extremity is concentrated specifically where the money questions are.

WHAT OUTSIDE RESEARCH ADDS

None of the above was built in isolation. A short literature check turned up two things worth flagging directly. First, cross-country survey research on "anchoring vignettes" — a technique built specifically to detect and correct for this kind of response-scale bias — confirms the general concern is real in principle: some cross-country variation in self-reported wellbeing genuinely is response style, not substance, which is exactly why this section exists rather than being skipped. Second, and more specific to Greece: independent academic and think-tank research (Katsikas, Karakitsios, Filinis & Petralias 2015, for the Crisis Observatory/ELIAMEP; Zambarloukou 2015, in *European Societies*) documents that Greece's elevated poverty profile predates the crisis for structural reasons — a comparatively thin welfare state with heavy reliance on family support — and that material poverty, especially once measured against a fixed rather than moving line, rose sharply after 2009. That's independent, Greece-specific confirmation of both halves of this section's own finding: a real pre-existing structural baseline (first test, above), and a real crisis-era material deterioration on top of it (Movement 3), not a reporting artifact for either.

Overall read **Greece entered the crisis with an unusually high reported level of financial difficulty, so a persistent country-specific reporting component cannot be excluded. But a stable reporting tendency cannot explain the subsequent deterioration and persistence by itself.** The right question isn't whether Greeks are culturally pessimistic — it's how much of the baseline and later movement remains once comparable material conditions and accumulated exposure (Movement 3) are already accounted for. On the evidence gathered here, most of it does.

Method: a six-test battery — coverage-audited pre-crisis ranking across three balanced panels, a formal difference-in-differences model with an event-study check for pre-existing trends, alternative crisis-start dates, a country-by-country placebo test, leave-one-country-out stability, a periphery-only comparison, and a synthetic-control check that failed its pre-period fit and is reported as a failure rather than as support — plus a standardized cross-indicator comparison and a specification-sensitivity check on Movement 3's residual trend. Full derivation, every test's exact numbers, and the underlying literature review are reproducible from `scripts/40_reporting_style_robustness.py`, `scripts/41_reporting_style_robustness_v2.py`, and `scripts/42_reporting_style_robustness_v3.py`, logged in full in `docs/publication_strategy.md`.

Where this fits in the research literature

None of this report's building blocks are being proposed here for the first time. What follows is a short, honest account of what other researchers have already established, how their work relates to this report's two questions, and where this report's own combination of pieces seems to be genuinely new.

THE MOVING-THRESHOLD PROBLEM: KNOWN, AND KNOWN FOR GREECE SPECIFICALLY

The core mechanism in Movement 2 — that a relative poverty line can understate a crisis because the line itself moves with the collapsing average — is not a new observation. Andriopoulou, Kanavitsa & Tsakloglou demonstrated it for Greece directly, using real household survey data for 2007–2016, and found an even larger effect than this report's approximation does once the two studies' year labels are properly aligned (see Movement 2 and Methods). Leventi & Matsaganis (2016), in an OECD working paper using the EUROMOD microsimulation model, independently confirm the same arithmetic: Greece's relative poverty rate barely moved across 2009–2014 (13.2% to 13.8%) despite a collapse in absolute living standards over the same span, precisely because the median income it's measured against fell in step. The OECD's own 2018 Economic Survey of Greece reaches the same conclusion from a different angle, noting that falling wages mechanically lowered the relative poverty threshold during the crisis. Goedemé, Decerf & Van den Bosch (2022), writing in the *Journal of European Social Policy*, proposed a new EU-wide poverty indicator specifically to fix this problem, and used Greece as their lead example of why it matters. So Movements 1–2 of this report is best read as building on and cross-checking that established finding with a different, subjective outcome variable — not discovering it.

SUBJECTIVE POVERTY AS A RESEARCH TOPIC: AN ACTIVE BUT DISTINCT LITERATURE

Subjective poverty measurement is its own established field, and it's worth being precise about a real difference in method: some studies (e.g. Želinský, Mysíková & Garner, 2022, *European Journal of Development Research*) build subjective poverty from a "minimum income question" (what income would your household need to get by), while this report — like Eurostat's own headline indicator — uses the "ability to make ends meet" question instead. They're related but not the same measure, and that EU-wide study using the other method also flags Greece and Bulgaria as exceptionally high, which is a useful independent cross-check even though the two studies aren't measuring identically the same thing. A related paper by some of the same authors (Želinský, Ng & Mysíková, 2020, *Economics Letters*) uses a statistical technique (the Youden index) to back out, from the same "make ends meet" responses this report uses, the income level that best separates households who call themselves poor from those who don't — a genuinely different question from this report's anchored-poverty measure (which asks how many people fall below a fixed *pre-crisis* living standard). It's a complementary way of turning subjective answers into an income figure, not an alternative version of Movement 2's construction — see the ideas list below for what a combined analysis could look like. Earlier work (Guagnano, Santarelli & Santini, 2016, *Social Indicators Research*) modelled subjective poverty across Europe against household and community characteristics, including social capital — a reminder that economic variables aren't the only candidate explanations in this literature, which bears on the open question in Movement 4.

WHAT MIGHT EXPLAIN A COUNTRY DOING WORSE THAN ITS ECONOMY PREDICTS

Movement 4's central puzzle — a country reporting persistently worse subjective outcomes than its measured economic conditions predict, even after controlling for a lot — has a close *parallel* in a completely different literature: the "happiness gap" between post-communist and Western European countries. Nikolova (2016, *European Journal of Political Economy*) found that gap wasn't fully explained by economic conditions either, and that institutional quality (the rule of law, specifically) explained a meaningful further share of it, on top of macroeconomics. That's not evidence about Greece — it's a study of a different set of countries and a different outcome (life satisfaction, not subjective poverty) — but it's a useful analogy for this report's own open residual, worth treating as a future exploratory hypothesis rather than the obvious next thing to test (household debt and welfare-transfer effectiveness were this report's own tests along similar lines — see the Answer to Question 2, the legacy section, for that result). There's also a specific, recent precedent for this report's housing-cost finding: Filandri, Pasqua & Tucci (2025, *Journal of European Social Policy*) link housing tenure and rent regulation to subjective poverty using EU-SILC data for young adults (18–34) across 24 European countries — a real, theoretically grounded precedent for treating housing tenure as a serious next variable, though their result is specific to that age group and hasn't been shown here to hold for the general Greek population. Movement 3's scarring finding has more direct grounding: Baldini, Gallo & Torricelli (2020, *Economia Politica*) find that a past income-scarcity spell continues to depress subjective make-ends-meet assessments up to two years later across Europe, even controlling for current income — the same mechanism, tested on a different (EU-SILC longitudinal, not cross-country) sample. Its pessimism finding is independently corroborated outside this report's own data entirely: a University of Macedonia survey for the Greek think tank diaNEOsis (2016) found 74% of respondents expected their own finances to worsen over the following year and 81.5% reported difficulty meeting basic household needs, using a different survey house, year, and question wording from anything else cited here. Nikolova's institutional-quality analogy, above, turns out to have direct evidence for Greece specifically, not just a parallel from a different country set: Ervasti, Kouvo & Venetoklis (2019, *Social Indicators Research*) and a 2026 OECD trust survey country profile, discussed fully in Movement 3, together describe institutional trust in Greece as a long-standing background weakness that the crisis deepened rather than created from a high starting point — context and mechanism for the pessimism finding, not an independent variable this report models or claims explains it away.

EXTERNAL CORROBORATION FROM OTHER GREEK DATA-JOURNALISM AND RESEARCH

Two independent, Eurostat/ELSTAT-sourced projects reach compatible conclusions using different comparison designs from this report's own. Mantés & Marinakis (2025), a report published via Greekonomics.gr with an accompanying open-source (MIT-licensed) codebase, benchmark Greece not against the EU average but against a "Bottom-10" group of ten EU countries that were poorer than Greece before 2008 and have since overtaken it — a comparator chosen specifically to separate genuine divergence from ordinary catch-up growth. Their income finding parallels Movement 3's own scarring result: Greek real disposable household income remained 18% below its 2008 level as of 2023, versus +11% for the EU27 average and +38% for the Bottom-10 group over the same span. On housing, they report overburden among Greece's below-median-income households running near 88–90% since 2012 — roughly three times the EU27/Bottom-10 rate on that same poverty-status-specific measure (not directly comparable to the legacy section's general-population housing figure, which is a different population base, but pointing the same direction). On healthcare, unmet medical need reached 12% of the Greek population in 2024, about five times the EU average — a variable outside anything this report tested directly, but a specific, concrete lead for future work given how large that gap is. Separately, a January 2026 diaNEOsis article on housing (summarizing an IOBE study led by Nikos Vettas) uses Eurostat's housing-cost-overburden definition directly and reports Greek households spending 35.5% of disposable income on housing in 2024 versus an EU average of 19.2%, with 42.8% of the population in some form of housing or utility arrears versus an EU average of 9.2% — that 42.8% figure exactly matches this report's own arrears rate for Greece (Movement 3), which is expected rather than a coincidence, since both draw on the same official Eurostat series, but is a useful independent sanity check regardless. diaNEOsis's own research portal also hosts a 2016/2017 extreme-poverty study (Matsaganis et al., Athens University of Economics and Business) built on a third, different poverty definition again — an absolute "basket of essential goods" income line, rather than AROP's relative 60%-of-median threshold or this report's own fixed-2008-anchor approximation (Movement 2) — worth noting as a reminder that "poverty line" is not a single settled concept even within Greek research specifically.

A STATISTICAL CAUTION WORTH NAMING

Cross-country regressions like the ones in Movement 4 — including in a lot of published research, not just here — usually treat each country as a fully independent observation. Claessens, Kyritsis & Atkinson (2023, *Nature Communications*) show this is often not a safe assumption: neighbouring or culturally-linked countries tend to move together in ways that standard statistics don't account for, which can make a finding look more solidly established than it is. This report's cross-country models use country-clustered standard errors, which helps but isn't the same thing as directly modelling geographic or cultural proximity, which is the stronger approach that paper tests. The right next step isn't necessarily a specific fix (their own comparison finds simpler corrections, like the one used here, are sometimes but not always adequate) — it's first checking whether this report's own residuals show spatial structure at all, then deciding how much correction that warrants. Worth flagging as a limitation this report shares with most work in this space, not something unique to it.

A CAUTION ABOUT USING AROP ALONE IN CRISIS CONTEXTS

The moving-threshold mechanism behind Movement 2 is well established in the poverty-measurement literature generally, not discovered here — what this report adds is a specific, concrete illustration of its scale. During Greece's crisis decade, a relative income-poverty rate like AROP moved far more slowly than actual living standards did, because the benchmark it's measured against was collapsing at the same time as the thing it measures. That is not a flaw specific to Greece or to this report's method; it is a structural property of any relative poverty line during a prolonged, broad-based macroeconomic collapse, and it generalizes more readily than any of this report's substantive Greek findings do. The practical implication: in a country undergoing sustained macroeconomic contraction, AROP reported alone can understate deterioration for years at a time. This report's own residual makes the same point from a different angle — even AROPE and a rich set of hardship controls leave real ground unexplained until a cumulative, exposure-based measure is added (Movement 3). The recommendation that follows is not "stop using AROP" — it remains a useful, standardized, comparable measure — but that in crisis or post-crisis contexts specifically, it is safest reported alongside a fixed-standard measure (an anchored poverty line, Movement 2), a broader official measure (AROPE), and some account of accumulated exposure over time, rather than read on its own.

WHAT, IF ANYTHING, IS NEW HERE

Given the above, this report's most defensible original contribution is the *combination*: tying the well-established moving-threshold mechanism specifically to the subjective "make ends meet" measure (rather than income poverty alone), testing that link with first-differencing and detrending rather than just raw correlation, checking whether it generalizes across the EU with a within-country panel, running the cross-country residual decomposition and nested leave-one-country-out predictions in Movement 4, and — going beyond current-year snapshots — showing that a cumulative, exposure-based measure of excess unemployment since 2009 narrows the residual that current-conditions models alone cannot (Movement 3). That specific combination — and the finding that Greece remains the largest cross-country gap of any EU state under every current-conditions specification tested, in-sample and out, until accumulated exposure is added — doesn't appear to already exist in the sources checked for this report. That's a claim worth treating cautiously rather than definitively: it comes from a targeted search, not a systematic literature review.

Full citations for every paper named above are in Methods.

Methods, robustness & limitations

Every series here originates from the Eurostat API dissemination endpoint — nothing hand-entered. Results reproduce both from the archived source-data snapshot and through the isolated `make reproduce` workflow, which exports a clean committed tree, re-fetches every raw input, rebuilds the pipeline, and verifies the published claims. A fresh pull is not expected to be byte-identical to the archived vintage because Eurostat revises historical data; 13 of 15 formerly orphaned inputs currently match byte-for-byte, while the two documented differences do not alter a headline model. This section gives the definitions and caveats behind every plain-language claim above.

SUBJECTIVE POVERTY — DEFINITION, EARLIEST DATA, AND OFFICIAL VERIFICATION

What's measured
Share of households answering "with difficulty" or "with great difficulty" to EU-SILC's "ability to make ends meet" question (`ilc_mdes09`, categories `DIF` + `GRT`). The underlying 6-point scale is unchanged in EU-SILC since 2003.

Verified against Eurostat's own headline product
Eurostat separately publishes an official "Subjective poverty" indicator, `ilc_sbjp01` (population-weighted, by age and sex), from 2010 onward. It matches this project's `ilc_mdes09`-derived household-weighted construction to within a few tenths of a point for Greece and the EU aggregate in every overlapping year (e.g. Greece 2024: 66.8% official vs. 66.7% here). The 2003–2009 portion of the series is this project's own same-methodology extension using the raw `ilc_mdes09` components, consistent with but not itself part of the official product.

Earliest comparable year

2003. In 2003–2004 only 6–13 countries had EU-SILC data yet, so Greece's EU rank in those two years is indicative only; the level (48.0%, 45.4%) doesn't depend on other countries and is robust.

AROP — DEFINITION, AND WHY IT ISN'T CALLED "OBJECTIVE POVERTY" HERE

What's measured

60% of national median equivalised disposable income (`ilc_li02` , `rskpovth=B_60` , `statinfo=MED_EI`). This report avoids the term "objective poverty" — Eurostat's own guidance is explicit that AROP measures income relative to other residents and does not by itself imply a low standard of living.

Reference-year mismatch

Greece's AROP rate for survey year *t* is generally based on income received in calendar year *t−1*, while the subjective question asks about the household's current situation. Both contemporaneous and 1-year-lagged correlations are reported for this reason.

AROPE — A DIFFERENT, BROADER INDICATOR, NOT A VARIANT OF AROP

What's measured

The share of people who are *either* AROP-poor, *or* severely materially and socially deprived, *or* living in a household with very low work intensity (a union of three conditions, not a weighted average) — Eurostat's EU2030 headline poverty-and-social-exclusion target indicator. Legacy series `ilc_peps01` (2003–2020) and revised series `ilc_peps01n` (2015–2025, using the new material-deprivation definition) are spliced at 2020/2021, the same break point documented for the deprivation series elsewhere in this report — not blended across the overlap.

Why it isn't a substitute for the anchored-poverty analysis

AROPE is a genuinely different measure, not another way of doing what Movement 2–3 do. Its severe-deprivation and low-work-intensity components aren't defined relative to a shifting national median the way AROP is, so it doesn't share AROP's specific "moving yardstick" mechanism — which is exactly why it was worth checking whether it tracks subjective poverty differently than AROP does. It does: correlation with subjective poverty is 0.48 in levels, 0.74 in year-over-year changes, and 0.90 once each series' own trend is removed, all well above AROP's 0.19 / 0.60 / 0.67 on the same three tests — and unlike the fixed-benchmark series in Movement 2, no approximation is involved.

EU ranking

Using the same EU-membership-by-year approach as AROP's ranking elsewhere in this report, Greece's AROPE rank has stayed within a 2nd–7th band across 2005–2025 — elevated, similar in shape to AROP's own 1st–10th range, and nowhere near subjective poverty's 1st-of-27 in every year since 2011.

ANCHORED POVERTY APPROXIMATION, VALIDATED WHERE POSSIBLE

Method

Eurostat's own anchored-poverty product (`ilc_li22`) is fixed at 2019, which misses the 2008–2013 collapse this report is about, so a 2008-anchored series is approximated instead. Each year, Eurostat publishes the poverty rate at four points — 40%, 50%, 60%, 70% of that year's median (`ilc_li02`) — and the euro income level each threshold corresponds to (`ilc_li01` , single-adult household). Treating income as approximately log-normally distributed, a probit-linear regression (poverty-rate percentile vs. ln(threshold)) is fit to those four points each year ($R^2 > 0.99$ in every year), then evaluated at the inflation-adjusted 2008 threshold (€6,480, chained forward/backward with Greek HICP).

Validation against the official 2019-anchored series

The identical method was used to independently reconstruct Eurostat's official 2019-anchored series (`ilc_li22`) for 2019–2025, then compared to the real published values. Mean absolute error: 1.22 percentage points. Maximum absolute error: 3.36 points (2022). Bias: the method runs +1.22 points high on average (a mild, consistent overestimate). Correlation with the official series: r=0.89 (p=0.007, n=7).

What the validation does and doesn't cover

Every validated year (2019–2025) required *zero* extrapolation — the anchor threshold stayed within the observed 40–70% range every time, because Greek incomes didn't move as far from 2019 as they did from 2008 during the crisis. The 2008-anchored estimates for 2012–2016 require real extrapolation beyond that observed range (peaking in 2016, when the anchor threshold sat noticeably below the lowest observed grid point that year). The validation therefore supports the method's general accuracy and its slight upward bias, but doesn't directly test the larger-extrapolation regime the most dramatic 2008-anchor estimates rely on. The crisis-era *direction and shape* (a sharp, sustained rise) is well supported independently by the directly-observed threshold collapse (Movement 2, no approximation involved); the exact anchored-poverty *levels* in 2012–2016 carry correspondingly more uncertainty than the validation numbers alone would suggest.

A second check, against real crisis-year microdata

Andriopoulou, Kanavitsa & Tsakloglou computed Greece's anchored poverty rate for 2007–2016 directly from household-level EU-SILC microdata (not a four-point approximation), anchored to 2007. Their result, for *income* year 2013: 48% of the population poor. Their years are labelled by income year; this report's series is labelled by EU-SILC survey year, which reflects income from the prior calendar year — so the aligned comparison is their income-year 2013 against this report's survey-year 2014 (40.6%), a gap of roughly 7–8 points. That gap should not be read as this report's approximation error specifically: the two exercises differ in several ways at once (household microdata vs. a four-point aggregate reconstruction, exact price-index and anchor-year choices, data vintage), so no single cause can be isolated. What the comparison does provide is real external corroboration, from independent microdata, of the same qualitative finding — and their estimate of the crisis-era effect is larger than this report's, not smaller. Andriopoulou, E., Kanavitsa, E., & Tsakloglou, P. Decomposing poverty in hard times: Greece 2007–2016. *Revue d'économie du développement* (2019); also LSE Hellenic Observatory GreeSE Paper No. 149 (2020), the version consulted here.

ROBUSTNESS TESTING: FIRST DIFFERENCES, DETRENDING, AND LEAD/LAG

Why

Two series that each have a strong multi-year trend (a crisis, then a recovery) can show a large raw correlation even without a real short-run relationship. First-differencing (year-over-year change vs. year-over-year change) and linear detrending (each series' deviation from its own trend line) both remove that shared trend and test whether the relationship survives.

Lead/lag, and why it's treated as secondary evidence

A −2..+2 year lead/lag scan was run on *first-differenced* series specifically (not levels) — a level-based lead/lag scan re-introduces the same shared-trend problem first-differencing exists to solve, since two smooth trending series will often stay correlated across a small shift regardless of any real short-run link. On first differences, unemployment shows the clearest, most stable pattern: correlation of roughly 0.5–0.86 across the window, with the same-year and prior-year values essentially tied at the top (0.859 and 0.855) rather than one clearly leading the other. That's directionally consistent with AROP's own known one-year reference lag, though the near-tie means this shouldn't be read as strong confirmation of it specifically. The anchored-poverty series and AROP itself are closer to each other on this stricter test than the raw-level comparison suggested (anchored poverty stays positive across the whole window; AROP turns slightly negative two years out) — a real but much more modest distinction than earlier framings implied, and reported here as descriptive timing evidence rather than as independent proof against a spurious relationship.

MULTIPLE-TESTING DISCIPLINE

Confirmatory vs. exploratory

This report uses four evidence tiers: **pre-planned confirmatory tests**; **exploratory screening**, corrected for multiple testing within each declared family; **post-selection robustness**, including placebo, leave-one-out, and nested procedures; and **descriptive corroboration**. Most later scorecard extensions were developed sequentially and are therefore exploratory or post-selection, not retroactively pre-specified. FDR controls false discoveries within a screening family; it does not turn model development into preregistration. Main claims require appropriate multiplicity control, out-of-sample stability, and a defensible substantive interpretation.

False-discovery-rate correction, applied to each exploratory family

Benjamini-Hochberg FDR correction was applied within each family of tests run together (not across the whole project, which would conflate tests answering different questions). **Movement 2's live correlation table (19 variables — the level, year-over-year, and trend-removed columns readers actually see) is corrected directly inside the script that produces it** (`10_robustness_correlations.py`), from full-precision p-values before any rounding, one correction per column: 17 of 19 survive on levels (AROP and income inequality do not — both are already described in the text as weak at the level and stronger on change); 16 of 19 survive year-over-year (severe material & social deprivation's newer measure, real wages, and headline HICP inflation do not); 17 of 19 survive trend-removed (severe material & social deprivation's newer measure and real wages do not; HICP inflation does survive this one, a genuine mixed case reported as such). Long-term unemployment, the strongest variable in the table, survives comfortably on every column (adjusted $p<0.0001$); youth unemployment and AROPE also survive comfortably throughout. Movement 3's gap-relation test (9 predictors): only real GDP survives, and only just (adjusted $p=0.050$); arrears, which looked significant before correction (raw $p=0.016$), does not (adjusted $p=0.070$) — consistent with the text's own read of it as likely Greece-specific rather than a general pattern. The acceleration/second-difference check (5 variables, not published in the main text): only material deprivation survives. Full adjusted values, with FDR columns included directly: `correlations_robustness.csv`, `fdr_sensitivity_vs_gap.csv`, `fdr_acceleration.csv`. **Separately, and superseded for the purpose of this table:** an earlier, differently-constructed 21-variable screen (contemporaneous level and one-year-lag correlations, `correlations.csv` / `fdr_correlations.csv`, from this project's original correlation pass before the year-over-year/detrended robustness columns existed) was also FDR-corrected at the time (16 of 21 survived) but is not the analysis behind the live table above and is kept only as an earlier exploratory record, not cited as governing any claim in the main text.

Labels used in the text

Where a finding's evidentiary status might not be obvious from context alone, it's tagged: Core finding for a confirmatory result central to answering this report's two research questions and, where relevant, to the scorecard; Descriptive check for a result that was tested directly and holds up on its own terms, but was deliberately not added to the scorecard (either because a formal model test wasn't feasible, or because it was tested and found redundant with a variable already in the model); Descriptive comparison, not a causal test for a real pattern in the data that wasn't tested with a regression or held to a significance threshold; Exploratory lead, not a core finding for a result from a multi-variable sweep that's suggestive but doesn't meet the same bar as the confirmatory tests. This isn't applied retroactively to every sentence in the report — it marks the specific places where the distinction is easy to miss.

CROSS-COUNTRY MODELS: PURCHASING POWER, NOT JUST CURRENCY

The cross-country models (Sections 7–9) use income adjusted for purchasing power (AIC per capita in PPS, `nama_10_pc`, `na_item=P41`), not raw currency — Eurostat's own guidance recommends this specifically for comparing household living standards *between* countries, since it removes cross-country price-level differences a currency figure doesn't. Chain-linked real EUR (the right choice for following *one* country through time) is kept for the Greece-only series in Sections 1–5. Swapping between the two bases changed Greece's average cross-country gap by only 0.2 points.

NESTED COMPARISONS: WHAT'S ADDED AT EACH STEP, AND ONE DELIBERATE CAUTION

Basic comparison: unemployment rate, purchasing-power-adjusted income, material hardship, AROP. **+ Housing:** adds housing cost overburden. **+ Bills/expenses:** adds arrears and inability to cover an unexpected expense. **+ Debt/transfers:** adds household debt-to-income and the poverty-reduction effect of social transfers. The "+ Bills/expenses" step is flagged deliberately: arrears and inability-to-cover-an-unexpected-expense are themselves close to financial-strain measures conceptually, not fully independent of the outcome being explained — so a shrinking gap at that step should be read as "closely related hardship indicators explain most of it," not as proof the basic comparison was simply too sparse. The leave-Greece-out test (the legacy section) uses the "+ Bills/expenses" specification as the most defensible middle ground between these two concerns.

SAMPLE STABILITY AND WITHIN- VS. BETWEEN-COUNTRY MODELS

Every cross-country comparison was rerun on a fixed set of 24 countries with complete data across all 10 years and all variables, instead of the full, changing EU membership used elsewhere; results were within 1–2 points in every case. Separately, Movement 2's within-country model (country + year fixed effects) and Movement 4's between-country model (year fixed effects only) answer genuinely different questions and are interpreted separately by design: country fixed effects remove each country's persistent baseline level, so that model cannot detect or speak to a Greek level effect at all.

EU AGGREGATES, COUNTRY RANKINGS, AND MATERIAL DEPRIVATION SERIES

No single Eurostat EU-aggregate code has constant country composition across 2003–2025; a priority fallback (current 27 states → earlier compositions → a changing-composition code) is used and recorded per year, which introduces methodological step-changes in the Greece–EU gap series on top of the economic ones (documented years: 2005, 2007, 2013).

Country rankings use EU membership as of each survey year (EU15 pre-2004 through EU27 post-Brexit from 2020), excluding EFTA/candidate countries that also report to Eurostat.

Two versions of "severe material deprivation" exist: a legacy 9-item measure (2003–2020) and a new 13-item measure (2021 onward, usable from ~2015). They are not directly comparable and are kept as separate series; the new measure's short overlap (10–11 years) is also why its correlation with subjective poverty collapsed after detrending.

VARIABLES TESTED BUT NOT CENTRAL TO THE STORY

Minimum wage (nominal, EUR/month) and **HICP inflation** (headline, food, housing & energy) were fetched and appear in the correlation table (Movement 2) — both came back weak and mostly not statistically significant, so they're reported there but not built on further. **Household consumption per capita** was used as a secondary income-side proxy in the same table (r=–0.70 in levels). **Household saving rate** was fetched and tested directly in the cross-country model as a candidate explanation for Movement 4's residual (Movement 3); it came back statistically null once financial-expectations pessimism was also in the model, despite Greece being a genuine outlier on it (2nd-lowest of 34 EU/candidate countries, 2015–2024 average). **Real wages**, previously flagged in this report as never fetched, now have been — see the dedicated entry below.

Income inequality (`ilc_di11`, S80/S20 income quintile share ratio; `ilc_di12`, Gini coefficient of equivalised disposable income) was fetched and tested as the final P2 checkpoint, specifically to check whether average-level hardship measures already in the model were hiding a worse-off tail. S80/S20 has full 27-country coverage, 2003–2024; Gini's age-broken-down series is shorter, 2014–2024 only — checked directly against Eurostat, not assumed. Greece is elevated but not extreme on either measure: 6th-highest of 27 in 2024, well behind Bulgaria, Lithuania, and Latvia. Greece's own S80/S20 has fallen since 2003 (6.38→5.27), including a substantial decline since its 2012 crisis-era peak (6.63). The level correlation with subjective poverty is weak and does not survive FDR correction (r=0.15, raw p=0.51, adjusted p=0.54) — the only variable in the correlation table with a non-significant level reading; first-difference and detrended correlations are stronger (r=0.64 and r=0.82 respectively) but the level result is the one that matters for the "is this a scorecard candidate" question. Added to Model C-LTU as a further predictor, it adds no independent explanatory power (R^2 0.914→0.915, Greece's out-of-sample gap moves from 3.9 to 6.0, not an improvement; coefficient not significant, p=0.50). **Not added to the scorecard, and not given its own subsection** — the useful finding here is the negative one: Greece's subjective-poverty gap is not mainly an inequality story. **Evidence role: ruled out** — tested directly and found not to carry the residual, distinct from the "descriptive, kept for context" role most of Movement 3's other checked-and-not-added variables play.

REAL WAGES: SOURCE, METHOD, AND WHY IT ISN'T A SCORECARD MODEL

Source and construction: nominal compensation per employee (`nama_10_lp_ulc` , `na_item=D1_SAL_PER` , EUR), deflated by the HICP price index (`prc_hicp_aind` , `coicop=CP00` , `unit=INX_A_AVG`), rebased so each country's own 2008 value equals 100. Full 27-country, 2000–2024 coverage — checked directly before use, and better coverage than several variables already in this report's own panel.

Multicollinearity check, done before considering it for the model: real wages correlate only modestly with the cross-country panel's existing income and scarring variables — PPS-adjusted income (r=–0.12), real GDP per capita (r=–0.12), the scarring stock (r=–0.33, Movement 3) — so it isn't simply duplicating them. It captures something conceptually different: labor compensation specifically, not total output or purchasing-power-adjusted consumption.

Model test, done and reported honestly: added to the legacy section's Model C as a candidate further predictor (same panel-regression setup: year fixed effects, country-clustered standard errors). R^2 moved from 0.892 to 0.893; the coefficient was not statistically significant (p=0.331); Greece's leave-one-out out-of-sample residual barely moved (11.59→11.18 points, rank unchanged at 1st of 27). Despite not being collinear with the existing variables, it doesn't add independent explanatory power to this specific cross-country comparison once they're already in the model — not added to the scorecard as a result. Kept in the report as a strong, separately-reported descriptive finding about Greece's own wage recovery (Movement 3), not folded into the regression.

WAGE-ADJUSTED PRICE PRESSURE: SOURCE, METHOD, AND LIMITS

Source and construction: Eurostat's comparative price level indices (`prc_ppp_ind_1` , `indic_ppp=PLI_EU27_2020`), EU27 average = 100, by COICOP consumption category. Wage benchmark: nominal compensation per employee (`nama_10_lp_ulc` , already used for real wages above), expressed as each country's value divided by the EU27 average, same 100-point scale. Wage-adjusted price pressure = price level ÷ wage level × 100, computed per country and category.

Coverage, checked directly before use: the "overall household consumption" category (`ppp_cat18=A01`) has full 27-country coverage back to 2000. The narrower categories used in the Movement 3 table (food, housing/water/electricity/gas, transport, restaurants/accommodation, information & communication) have full 27-country coverage for only 2022–2024 — Eurostat does not publish the detailed category breakdown further back. This is a real feasibility constraint, not an oversight: it rules out a cross-country panel-regression model test for the category-level detail (too little time variation for the year-fixed-effects setup used elsewhere in this report).

What this is not: a household-budget survey or actual supermarket receipts. Price level indices are harmonized, category-wide comparisons built for cross-country comparability, not a measurement of what any specific Greek household spends on any specific basket of goods. The wage-adjusted pressure figure is a transparent ratio of two published Eurostat indices, not a constructed cost-of-living or affordability index with its own independent methodology — deliberately labeled "price pressure," not "affordability," for that reason.

Correlation and model role: only the overall-consumption series has enough years to check against subjective poverty. Level correlation r=0.696 (p=0.0003); first-difference r=–0.063 (not significant) — the same level-only pattern already documented for real wages. No cross-country model test was run given the coverage constraint above; not added to the scorecard.

HOUSING TENURE: SOURCE, METHOD, AND WHY IT'S DESCRIPTIVE DEPTH, NOT A NEW MODEL VARIABLE

Source and construction: housing cost overburden by tenure status (`ilc_lvho07c` : mortgage-free owners, owners with a mortgage, market-rate renters, reduced-rate/free renters) and the tenure distribution itself (`ilc_lvho02` : share owning vs. renting, with or without a mortgage). Full 27-country coverage on both, 15–22 years depending on country. The "TOTAL" row of `ilc_lvho07c` exactly matches the `housing_cost_overburden` variable already used throughout Sections 7–11 and in the cross-country model — checked directly, not assumed.

Data-quality note: the renter-specific subseries (market-rate and reduced-rate renters) show implausibly large year-to-year swings in Greece's historical data before roughly 2021 (for example, market-rate renter overburden reported at 23.6% in 2007 and 87.5% in 2014) — consistent with a small underlying renter subsample in a historically ownership-dominated country, not a real one-year shift in housing conditions. The 2022–2024 window is stable and is what this report relies on for the by-tenure comparison; the full historical series was not used for any trend claim.

Multicollinearity check and model test, done before deciding not to add it: mortgage-free-owner overburden correlates at r=0.91 (panel-wide) and r=0.88 (Greece-only) with the `housing_cost_overburden` variable already in Model C-LTU — expected, since it's a subcomponent of that total by construction. Added to Model C-LTU as a further predictor, R^2 moves from 0.914 to 0.917, Greece's out-of-sample gap moves from 3.9 to 5.1 (not an improvement), and the coefficient is not statistically significant (p=0.141). **Not added to the scorecard. Evidence role:** material-scarring context, the same role real wages and wage-adjusted pricing play — the housing-cost channel is already captured in the scorecard via the broader overburden variable; this entry adds texture to why it's so severe, not new statistical weight.

Correlation, reported separately from the model test: Greece-only over time, mortgage-free-owner overburden vs. subjective poverty, r=0.90 (p<0.0001, n=18); cross-country, single year (2024), r=0.66 (p=0.0002, n=27). Both descriptive, not treated as evidence of a causal ownership-specific channel.

LONG-TERM UNEMPLOYMENT: SOURCE, METHOD, AND WHY IT REPLACES RATHER THAN ADDS TO HEADLINE UNEMPLOYMENT

Source and construction: Eurostat's long-term unemployment series (`une_ltu_a` , `indic_em=LTU` , unemployed 12 months or more as a percentage of the active population, ages 15–74). Full 27-country coverage, 2009–2024 (Eurostat does not publish this series further back).

Multicollinearity check, done before deciding how to use it: long-term unemployment correlates strongly with headline unemployment — r=0.91 across the full 27-country, 16-year panel; r=0.998 for Greece's own time series alone, since by the mid-2010s the large majority of Greece's unemployed had been out of work a year or more. Adding both to the same regression together produces a variance inflation factor of roughly 8–9 for each — a real, non-trivial collinearity level, though below the more extreme thresholds sometimes used to disqualify a variable outright (commonly VIF>10).

Two specifications tested, one preferred: (1) *added* alongside headline unemployment (Model C+LTU): R^2 rises to 0.919, Greece's out-of-sample gap falls to 2.3, but headline unemployment's own coefficient flips to a statistically insignificant negative value (p=0.093) — a textbook collinearity symptom, reported as a robustness check only, not the preferred model. (2) long-term unemployment *replacing* headline unemployment (Model C-LTU): R^2 rises to 0.914, Greece's out-of-sample gap falls to 3.9 (rank 1st→6th of 27), with clean, stable coefficients throughout. This report treats (2) as the preferred specification. Long-term unemployment's own coefficient in Model C-LTU is stable across all 27 leave-one-country-out refits (range 2.56–3.44, always p<0.001, including the refit excluding Greece itself) — evidence this is a robust relationship, not an artifact of Greece's own data or of the collinearity with headline unemployment.

Interaction with the other Movement 3 variables, tested directly: adding the scarring-stock variable (below) on top of Model C-LTU changes the fit negligibly (R^2 unchanged at 0.914) and its own coefficient is not significant (p=0.605) — long-term unemployment appears to subsume most of what the scarring-stock variable was capturing. Adding financial expectations and saving rate (below) on top of Model C-LTU still improves the fit (R^2 0.920) and Greece's gap (down to −1.75, meaning slightly over-predicted, rank 19th of 27) — but financial expectations' own coefficient weakens from p=0.021 (with headline unemployment) to p=0.062 (with long-term unemployment), and this combined three-variable specification is the most heavily stacked one tested. It is reported in Movement 3 as a directional finding (some overlap exists) rather than a headline number, both because financial expectations was already the most heavily caveated variable in this report and because stacking several plausibly-related "scarring-adjacent" variables in a smaller sample (n=265) risks looking more conclusive than it is.

Correlation robustness: long-term unemployment is the strongest and most consistent variable in the whole correlation table — level r=0.933, first-difference r=0.922, detrended r=0.859, all p<0.0001, and it survives Benjamini-Hochberg FDR correction comfortably. Most other variables in this report's correlation table weaken substantially once detrended or first-differenced; long-term unemployment does not.

Full model outputs: `model_scorecard_ltu.csv`, `scorecard_loo_C_baseline.csv`, `scorecard_loo_C_LTU_swap.csv`, `scorecard_loo_C_LTU_add.csv`, `scorecard_loo_C_LTU_plus_scarring.csv`, `scorecard_loo_C_LTU_plus_expectations.csv`.

CUMULATIVE EXCESS UNEMPLOYMENT: SOURCE, CONSTRUCTION, AND EVIDENTIARY STATUS

Construction: for each country, a 2009 baseline unemployment rate (the earliest year with near-complete 27-country coverage for both headline and long-term unemployment, confirmed by direct coverage check rather than assumed). For every subsequent year, that year's unemployment rate minus the baseline rate, floored at zero (a year at or below baseline contributes nothing, rather than offsetting a later bad year), summed cumulatively through 2024. This produces a running total of "excess unemployment-years" carried since the crisis began, distinct from the current-year snapshot headline or long-term unemployment rates already in the model. The equivalent construction was also tested for long-term unemployment, GDP per capita (two baselines: fixed-2008 and each country's own rolling historical peak), real wages (same two baselines), and the AROP poverty threshold itself (fetched in national currency to avoid exchange-rate contamination, deflated by each country's own HICP).

What was tested, and what survived: every cumulative candidate was added, one at a time, to the Model C-LTU baseline. Cumulative excess unemployment was the strongest (R^2 0.914→0.930, Greece's out-of-sample gap 3.9→–0.8, both legacy and superseded). A duration/direction battery on GDP and wages (years below 2008, years below own peak, longest continuous shortfall streak, cumulative negative-growth years) found one further significant candidate — years spent below Greece's own 2008 real-wage level ($p=0.0045$) — and several non-significant ones (all GDP-based variants, $p=0.13$–0.89). A cumulative shortfall in the AROP poverty threshold itself, despite being the most narratively anticipated candidate given Movement 2's moving-threshold finding, came back statistically null ($p=0.291$).

Replacement test: to check this isn't long-term unemployment relabeled, cumulative excess unemployment was tested as a *replacement* for long-term unemployment as well as an addition. Replacement performs worse (R^2 0.911, Greece's out-of-sample gap 4.3) than either the long-term-unemployment baseline (R^2 0.914, gap 3.9) or the combined model (R^2 0.930, gap –0.8 — legacy, superseded) — evidence this is a genuine additional layer, not a substitute. Variance inflation factors in the combined model stay reasonable (maximum 4.5).

Stability checks: refitting 27 times, once with each country excluded in turn, cumulative excess unemployment's coefficient stays positive and significant throughout (range 0.134–0.197, no sign flips), including the refit that excludes Greece itself (coefficient 0.164, $p=0.0064$). The wage-duration variable passes the same check (range 0.31–0.54, always significant; excluding Greece: 0.31, $p=0.0036$). A year-by-year check of the final model's residual for Greece (fit excluding Greece) shows small, mostly negative-to-near-zero residuals for nine of ten years 2015–2024 (–3.25 to +1.01), with one exception (2021, +4.40, plausibly pandemic-related) — not a single-year artifact.

Why the wage-duration variable is kept as supporting evidence, not a further scorecard layer: combined with cumulative excess unemployment, both variables remain statistically significant (R^2 0.931, gap –1.1), but within Greece's own 2003–2025 time series the two are highly correlated (0.95, versus 0.54 across the full 27-country panel) — too entangled in Greece's specific case to treat credibly as two independent mechanisms, even though the panel-wide relationship supports both as real.

Multiple-testing correction across the full cumulative/duration family: the original 8-candidate cumulative battery and the 10-candidate duration/direction battery were screened together as one exploratory family (18 tests total, the same discipline applied elsewhere in this report), then checked against Benjamini-Hochberg false-discovery-rate correction — computed inside `scripts/38_cumulative_hardship.py` itself from full-precision p-values, not as a separate one-off calculation. Cumulative excess unemployment survives comfortably (raw p=0.00013, FDR-adjusted p=0.0024). The wage-duration variable survives, but narrowly (raw p=0.0045, FDR-adjusted p=0.0408, just inside the conventional 0.05 threshold). None of the other 16 candidates survives correction (adjusted p≥0.25 in every case). Full correction table: `cumulative_hardship_fdr_correction.csv`.

A selection-sensitivity check, reported honestly rather than smoothed over: cumulative excess unemployment was picked as the preferred candidate using the full 27-country panel, Greece included — the way variable screening is legitimately done in this report, since Greece's data is a real part of the historical record. But that means the leave-one-country-out checks above test whether the *already-chosen* variable's coefficient is stable without Greece, not whether Greece's own data point influenced *which* variable got chosen in the first place. Checked directly: rerunning the full 18-candidate screening with Greece dropped from the panel entirely (not just held out for evaluation, excluded from the fitting altogether), the wage-duration measure edges ahead of cumulative excess unemployment (p=0.0036 vs. p=0.0064, both still highly significant either way). The two candidates that matter don't change — the same two survive FDR correction whether or not Greece is in the screening panel — but which of them ranks first is closer than the headline framing suggests. Read as two closely related, mutually reinforcing measures of sustained hardship (they correlate at 0.95 within Greece's own data, as noted above), not as one uniquely, unambiguously identified mechanism with the other as a clear runner-up. Cumulative excess unemployment is kept as the headline construction here because it has the more direct interpretation (a labor-market variable, not one step removed via wages) and remains the stronger FDR survivor with Greece properly included in screening. Full comparison: `cumulative_hardship_selection_excl_greece.csv`. **The stronger version of this check has now also been run** (`scripts/44_nested_selection_validation.py`): the complete 18-candidate screening repeated independently inside every one of the 27 leave-one-country-out folds — each country dropped from the screening entirely, not merely held out for evaluation. Cumulative excess unemployment is selected first in 25 of 27 folds; the Greece-excluded fold selects the wage-duration measure (the sensitivity already documented above), and one further fold (Finland excluded) selects a closely related GDP-duration measure, with cumulative excess unemployment still ranked 3rd there and significant in every single fold (worst-case p=0.0064). Selection is stable, and the two exceptions both land on near-neighbor duration/exposure constructions rather than on anything outside the family; in 26 of 27 folds the fold's own winner also survives FDR correction applied within that fold. **The procedure was then scored end to end**, which is the stricter test: in each fold, that fold's *own* selected variable is fitted on the 26 training countries and used to predict the held-out 27th, which took part in neither selection nor fitting. Greece's residual under that fully nested procedure is **+2.70 points** *(legacy, superseded by the objective-only headline)* (mean absolute error 4.51), ranking 19th of 27 — **eight EU countries are predicted less accurately than Greece** by their own fold's model. That number, not the –0.8 of the fixed final specification — both superseded by the objective-only headline in Movement 4 — is the one to cite when the claim is about the whole select-then-fit procedure rather than one already-chosen variable: –0.8, a legacy superseded figure, comes from a model whose variable was picked with Greece in the screening panel, while +2.70 also pays the cost of that selection step. Both belong to the superseded, proximity-sensitive family, and +2.70 is the more conservative estimate *within that family* — not the project's more defensible current result, which is the +6.93 from the objective-only specification in Movement 4. Neither figure supports the statement that Greece stops being a cross-country outlier: under the current specification Greece remains third of 27. Full per-fold results: `nested_selection_validation_folds.csv`, `nested_selection_validation_detail.csv`.

Testing the "permanent accumulation" assumption directly: because the cumulative sum can only grow, on its own it can't distinguish genuine undiminished accumulated exposure from simply encoding "years since the crisis began." Tested five alternative constructions from the same underlying data, each of which *can* fall over time: trailing 3-, 5-, and 10-year rolling sums, and two exponentially-decayed running sums (20%/year and 10%/year decay). A 3-year window is too short to capture the effect at all (not statistically significant); a 5-year window is borderline. But a 10-year rolling window fits at least as well as the permanent since-2009 sum — slightly better, in fact (R^2 0.932 vs. 0.930, Greece's out-of-sample gap –0.47 vs. –0.82, both legacy superseded figures, both very significant) — and both decay-rate versions also fit strongly. The honest reading: this is evidence the association is stable across constructions and not a short-term artifact, but it's better described as sustained excess unemployment over roughly the past decade than as literal permanent, never-fading accumulation since one specific baseline year — a claim now backed by several related constructions converging on the same conclusion, not resting on a single, somewhat arbitrary choice. Full battery: `cumulative_hardship_rolling_decay_battery.csv`.

Not without precedent: extending an idea the EU already applies at the individual level. Eurostat's own persistent-poverty indicator counts someone poor only if they're AROP-poor in the current year *and* in at least two of the previous three — because a single-year snapshot is known to miss how widespread and durable poverty experience actually is. A 2026 European Commission study of EU-SILC longitudinal data (2011–2022) reaches the same conclusion and explicitly recommends dynamic measures for policy design. Cumulative excess unemployment applies the same logic one level up — though the mechanism that literature describes operates at the household level and is **not tested anywhere in this report**, which observes country aggregates only. Instead of asking whether a household was poor across several recent years, it asks whether a country's labor market stayed depressed relative to its own baseline across several recent years. At the macro level, this also mirrors the "hysteresis" mechanism the IMF has documented for the euro area — prolonged unemployment eroding skills and detaching workers from the labor force in ways that compound rather than reset year to year.

What this variable can and can't show: a level-of-analysis caveat. The individual-level literature on unemployment scarring is substantial: past unemployment continues to depress life satisfaction even after re-employment (a 15-year German panel study), each additional six-month unemployment spell predicts lower later-life satisfaction across 14 European countries even controlling for current conditions, and cumulative unemployment before age 30 predicts lower wellbeing at 30 in British cohort data. All three measure individual employment histories and individual outcomes. This report's cumulative excess-unemployment variable is a country-year aggregate, and its correlation with Greece's aggregate subjective-poverty rate is not direct evidence that the specific Greek households driving that rate personally experienced sustained joblessness. The aggregate result is consistent with the same underlying mechanism documented at the individual level, but this report doesn't observe individual employment histories, and the two kinds of evidence shouldn't be conflated.

On the interpretation of the negative out-of-sample residual (legacy, superseded): a residual of −0.8 means the richest specification tested slightly overpredicts Greece's reported hardship, not that Greek households are secretly less pessimistic than their conditions warrant. This is a property of this specific, richest specification (Model C-LTU plus cumulative excess unemployment), not the report's baseline finding, and should not be read as "the gap is more than 100% explained" — a phrasing this report deliberately avoids.

Full derivation, every candidate tested, and all robustness checks: `scripts/38_cumulative_hardship.py`; output tables `cumulative_hardship_checkpoint.csv`, `cumulative_hardship_duration_battery.csv`, `cumulative_hardship_replacement_test.csv`, `cumulative_hardship_loo_stability_cum_excess_unemployment.csv`, `cumulative_hardship_loo_stability_wage_years_below_2008.csv`, `cumulative_hardship_final_model_year_by_year.csv`, `cumulative_hardship_stage1_individual.csv`, `cumulative_hardship_stage2_bridge.csv`, `cumulative_hardship_fdr_correction.csv`, `cumulative_hardship_selection_excl_greece.csv`, `cumulative_hardship_rolling_decay_battery.csv`; full narrative log in `docs/publication_strategy.md`.

YOUTH UNEMPLOYMENT: CHECKED, STRONGLY CORRELATED, AND NOT A SCORECARD MODEL

Source and construction: Eurostat's youth unemployment rate (`une_rt_a`, `age=Y15-24`, `unit=PC_ACT`) — the share of the youth *labor force* that is unemployed, not the share of the youth *population* (a different, lower-looking Eurostat indicator, `yth_empl_090`, that this report does not use). Full 27-country coverage, 2009–2024, the same window as headline and long-term unemployment.

Descriptive: Greece peaked at 59.2% in 2013 and fell to 22.5% by 2024. Unlike almost every other variable in this report, Greece is no longer the extreme outlier on this specific measure: Spain overtook it (26.5% in 2024), and Greece now ranks 4th-highest in the EU (behind Spain, Sweden, and Romania) rather than 1st.

Correlation with subjective poverty: level r=0.71 (p=0.002), first-difference r=0.79 (p=0.0005), detrended r=0.81 (p=0.0002) — genuinely strong, and it survives FDR correction comfortably (adjusted p=0.0028), but weaker at every time scale than long-term unemployment's r=0.93/0.92/0.86.

Overlap, checked directly: extremely high collinearity with both headline unemployment (r=0.92 panel-wide, r=0.99 for Greece's own series) and long-term unemployment (r=0.85 panel-wide, r=0.99 for Greece's own series) — higher overlap than long-term unemployment has with headline unemployment.

Model test: replacing headline unemployment with youth unemployment in Model C leaves Greece the largest out-of-sample outlier in the EU (R^2 0.899, gap 10.8, rank unchanged at 1st of 27) — a much smaller improvement than long-term unemployment's swap (R^2 0.914, gap 3.9, rank 6th). Added on top of Model C-LTU instead (robustness), it changes almost nothing (R^2 unchanged at 0.914, gap 3.5, rank unchanged at 6th) and its own coefficient is not statistically significant (p=0.815, variance inflation factor 4.7) — once long-term unemployment is already in the model, youth unemployment adds no further independent explanatory power. **Not added to the scorecard as a result.**

Where it's used instead: as supporting descriptive context for the migration finding (Movement 3) — Greece's youth unemployment peak (2013) and its emigration peak (2012) fall within a year of each other, a plausible but not causally tested push-factor connection.

MOVEMENT 3 VARIABLES: SCARRING STOCK AND FINANCIAL EXPECTATIONS

Distance below own peak (scarring stock): built from real GDP per capita (`sdg_08_10`), 2008–2024. For each country and year, its running historical maximum up to and including that year, minus its actual value that year, as a percentage of the peak. Added to the the legacy section/9 "+ Bills/expenses" specification as a further predictor (year fixed effects, country-clustered standard errors), the same panel-regression setup used throughout Movement 4. This differs from an earlier, purely cross-sectional check in this project (a single peak-to-trough decline number correlated against average residuals, kept in the codebase as an exploratory script) — the Movement 3 version is time-varying and enters the regression directly.

Financial expectations: Eurostat's household consumer survey (`ei_bsco_m` , indicator `BS-FS-NY`), "financial situation of household over next 12 months," a balance statistic (net % expecting improvement minus deterioration). **Household saving rate:** `tec00131` , gross saving as a share of disposable income. Both merged into the same panel-regression setup as the scarring variable, added together as a further specification beyond the legacy section/9's richest model.

Recovery trajectory: also built from `sdg_08_10` , 2008–2024. Two measures: current distance below each country's 2008–2024 peak (as above), and separately, each country's worst historical drawdown (maximum-peak-to-later-trough decline) and whether/when a later year's value returned to the peak that preceded it. Not added to any regression — reported descriptively, cross-checked against the already-published scarring-stock numbers (exact match on the current-distance-below-peak figures) as a consistency check on the new calculation.

Net migration of Greek nationals: Eurostat's migration-flow tables, `migr_emi1ctz` (emigration) and `migr_imm1ctz` (immigration), both filtered to `citizen=NAT` (the reporting country's own nationals only) and `agedef=COMPLET`, 2008–2024, all 27 EU countries. Population denominator for the rate calculation: `demo_pjan`. Eurostat's headline "net migration rate" product (`demo_gind`, indicator `MIGTRT`) was checked first and rejected: it is officially "net migration plus statistical adjustment," a residual that absorbs census and population-register revisions, and for Greece it shows a strongly *positive* net migration every year 2011–2024 — directly contradicting Greece's own falling population and the well-documented emigration pattern, a clear sign it's dominated by statistical noise for this country rather than real migration. The flow tables used instead give a coherent, plausible pattern and all 27 EU countries report into them. Age and education breakdowns are not available for this project's own flow data restricted to Greek nationals specifically (only the aggregate total); where an age or education claim appears in the text, it is explicitly attributed to the OECD/EKT census-based source instead, not to this table.

Institutional trust: reviewed, not modeled. Eurostat's entire holdings on trust in institutions and public services amount to a single dataset (`ilc_pw03b`) covering exactly one year, 2013 — checked directly against Eurostat's own catalogue, not assumed. That single cross-sectional year cannot support a time-varying regression variable or a trend claim, so none was built. (Eurostat's only *ongoing* annual trust series, `ilc_pw03` / `ilc_pw04`, measures interpersonal trust — trust in other people — a different construct from institutional trust, and not substituted for it here.) Broader Eurobarometer/European Social Survey/OECD sources support the interpretive discussion in Movement 3 and the literature section, but are not integrated into this report's reproducible Eurostat-only panel model in this version — flagged as a candidate for a future version that expands beyond Eurostat's own API.

WHAT WAS DELIBERATELY EXCLUDED

The pre-2003 European Community Household Panel was not merged into any series (different sample design, differently-worded question) — 2003 is a hard start throughout. Unemployment data at the exact age band used here has no disseminated Greek values before 2009, despite other series running back to 2000. All data reflects the current Eurostat vintage as fetched; no attempt was made to reconstruct as-first-published values.

REMAINING LIMITATIONS: WHAT'S ADDRESSED BY DESIGN, AND WHAT'S STRUCTURAL

Addressed through the checkpoint's own design: the Stage 1/Stage 2 structure behind Movement 3's cumulative-hardship findings exists specifically to prevent two common overclaiming risks. Stage 1 tests every candidate alone against the same simple baseline, so a variable's apparent strength can't be an artifact of which other variables happened to already be in the model. Stage 2 shows the gap-closing sequence is a deliberate layered comparison, not an arbitrarily stacked one.

Reduced through framing, not eliminated: the AROP/AROPE hierarchy (Sections 1, 7) keeps this report's primary claims anchored to a single, well-defined measure rather than switching between two whenever convenient. Variables conceptually close to the outcome itself (arrears, unexpected-expense capacity, financial expectations) are labeled as such everywhere they appear, not just once. Short-coverage series (institutional trust, granular price categories) carry explicit coverage notes rather than being extrapolated. Migration and institutional trust are kept as contextual scarring, described but never modeled as regression covariates.

Structural, not reducible at this level of data: every result in this report is drawn from aggregate, country-level Eurostat series, not EU-SILC household microdata — AROPE's three components overlap at the household level in ways this report cannot observe or decompose. All relationships reported are associational, not causal. Household balance sheets, informal income, savings buffers, and family support networks are only partially observed, if at all, through the variables tested. Subjective poverty is self-reported and may reflect shifting expectations or social norms as well as material conditions. Greece's crisis is, by the literature's own account, distinctive in scale and duration (Movement 3), so the AROP-in-crisis-contexts caution above is expected to generalize more readily than any single substantive Greek finding in this report.

DATA APPENDIX

Every number in this report traces back to a cleaned, reproducible table, not just to charts. The core Greece–EU annual series is `master_table.csv`; the full merged dataset is `analysis_dataset.csv`; model and robustness outputs are stored in separate named CSVs. The numbered scripts, dependency-aware Makefile, isolated `make reproduce` target, and automated verification checks rebuild and audit the complete pipeline. Fresh Eurostat revisions can change historical source values, so the archived vintage remains the reference for exact quoted numbers.

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Data & methods, in brief. All series pulled from the Eurostat API dissemination endpoint. Subjective poverty = "difficulty" + "great difficulty" making ends meet (`ilc_mdes09`, cross-checked against official `ilc_sbjp01`). AROP = 60% of national median equivalised income (`ilc_li02`) — a relative-income measure, not treated as a direct living-standards indicator. Cross-country models use purchasing-power-adjusted income. Correlation and regression are descriptive, not causal. See "Methods, robustness & limitations" above for every definition and caveat.